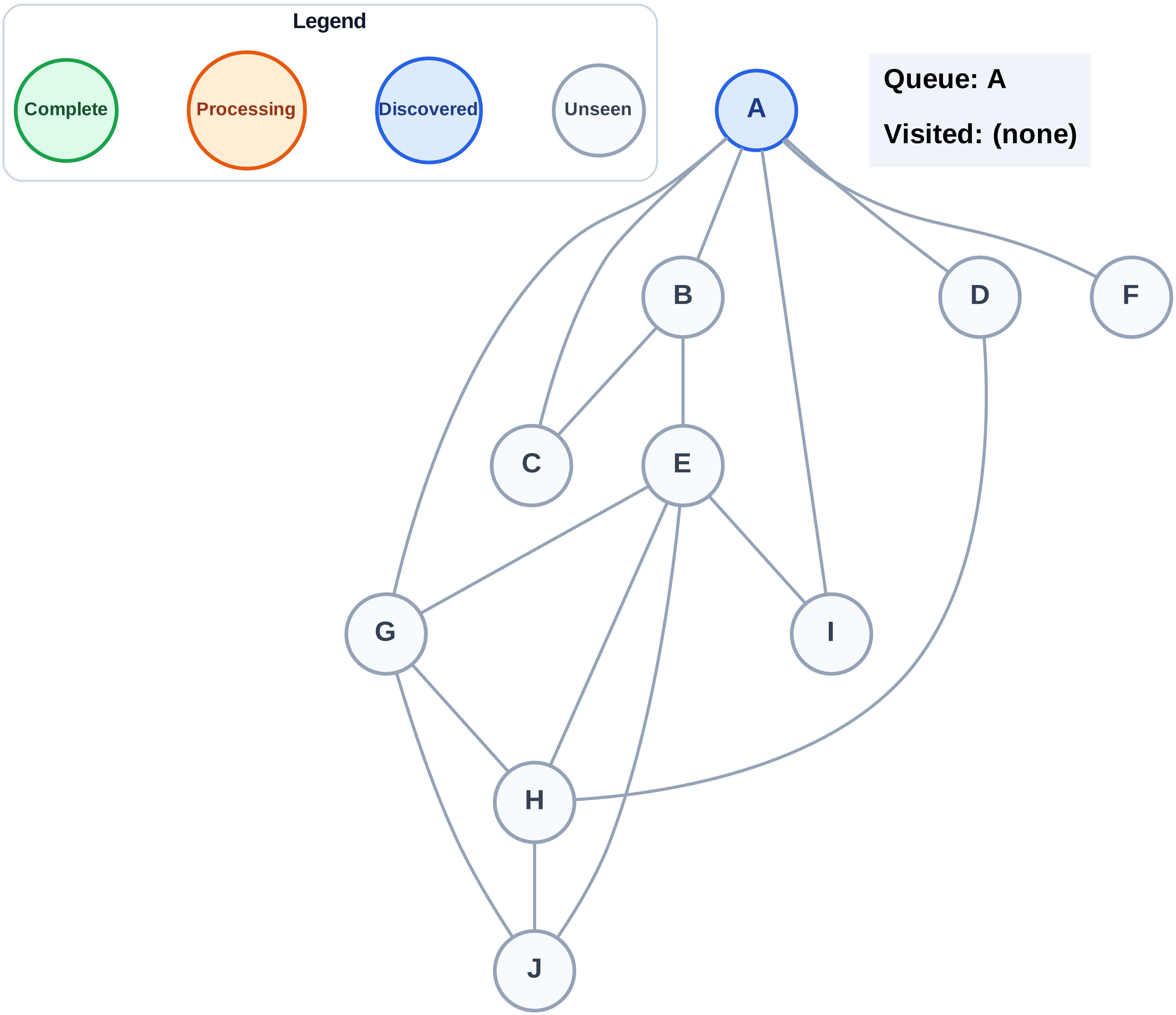
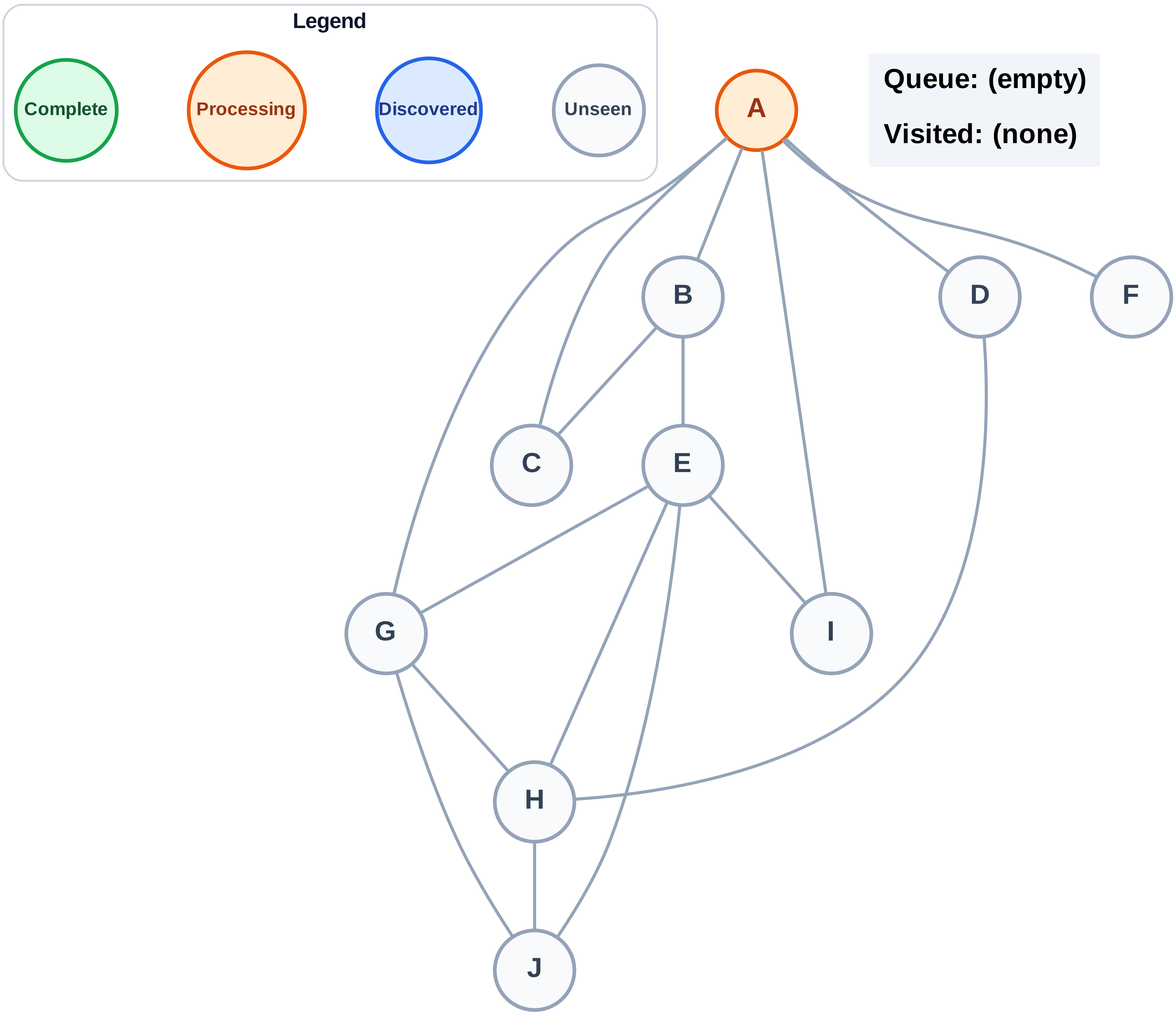
BFS Traversal

Step 1: Start at A



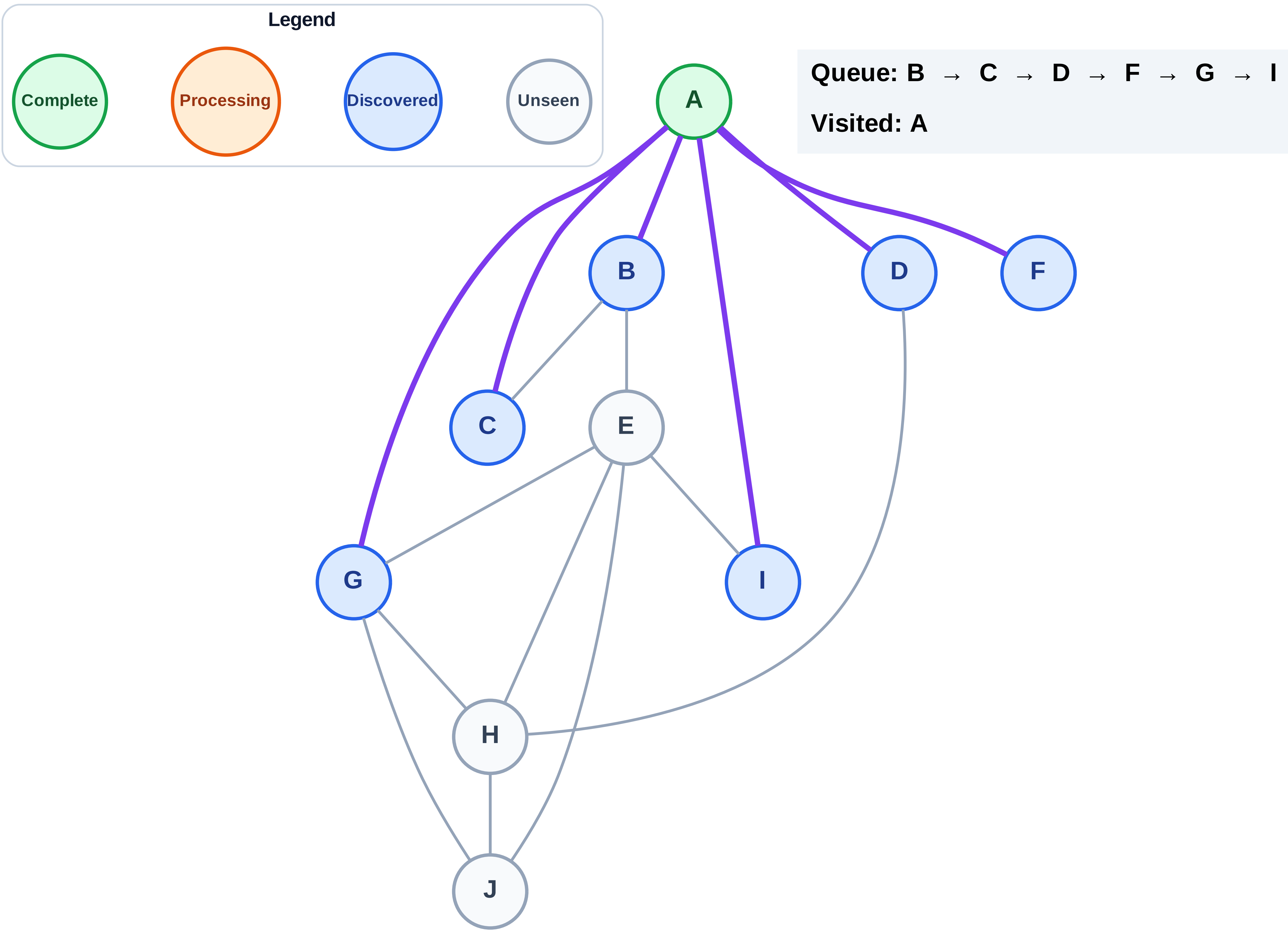
BFS Traversal

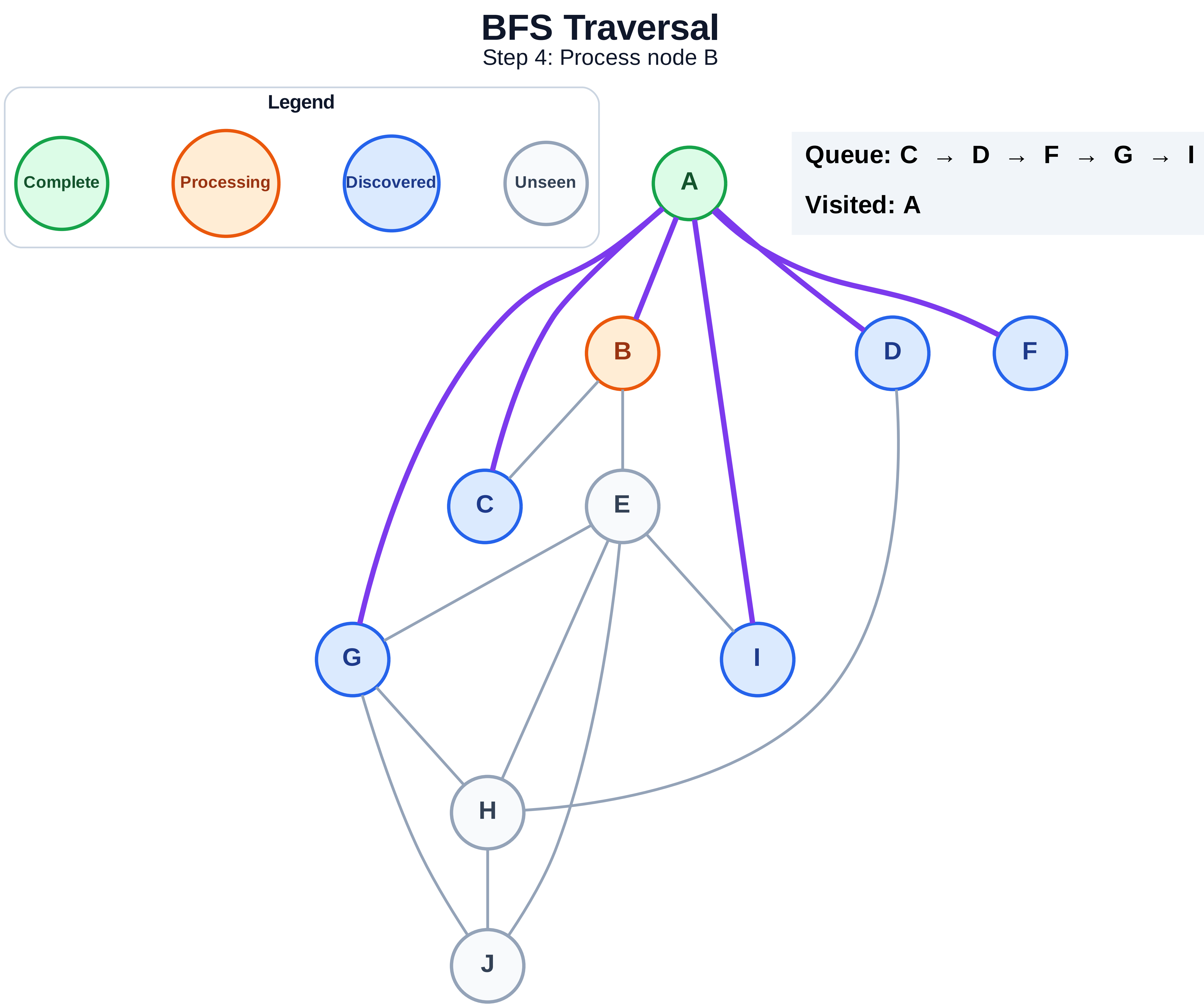
Step 2: Process node A



BFS Traversal

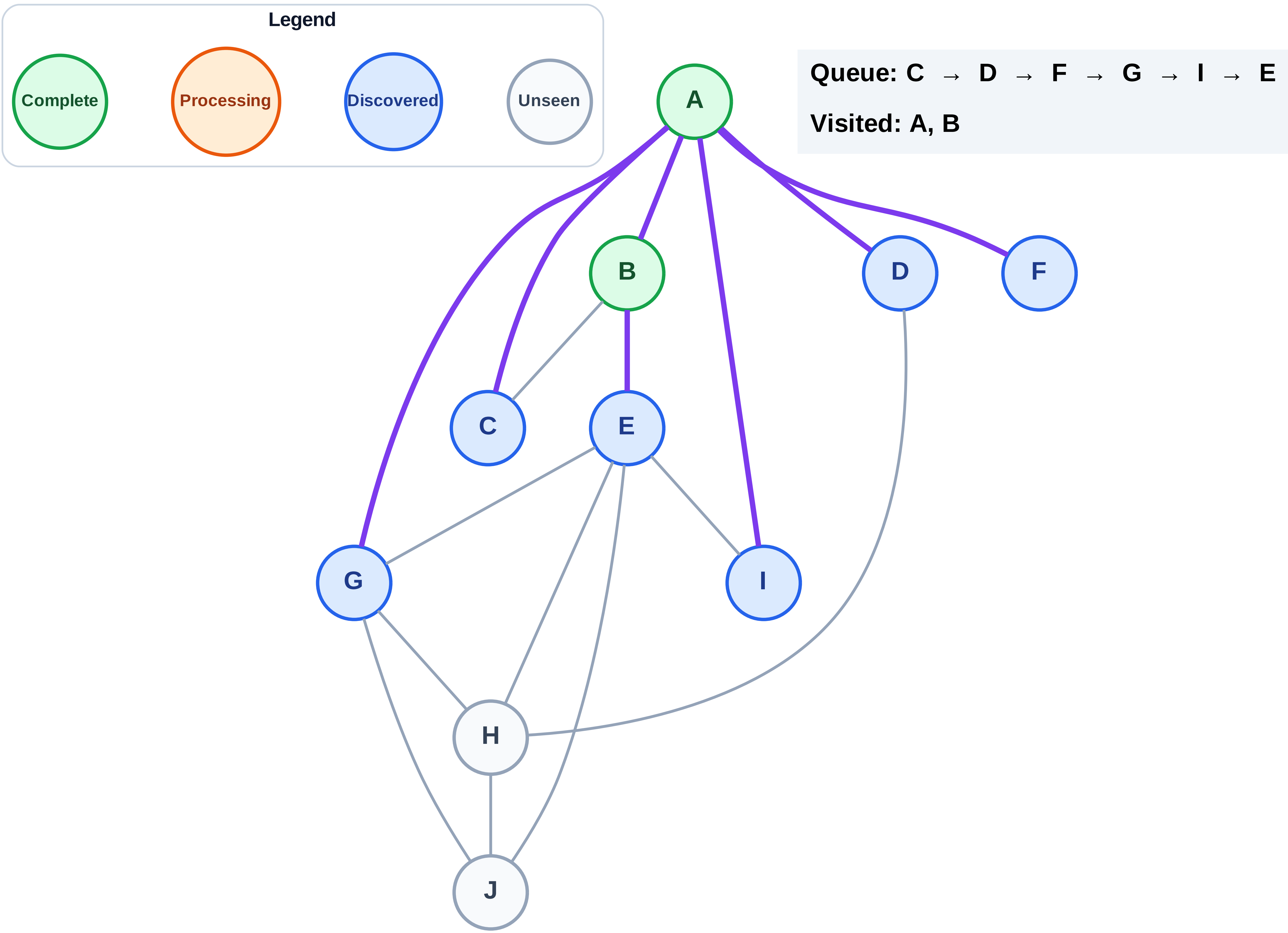
Step 3: Discover B, C, D, F, G, I from A




BFS Traversal
Step 4: Process node B
Legend
Complete
Processing
Discovered
Unseen
Queue: C → D → F → G → I
Visited: A
A
B
C
D
E
F
G
H
I
J

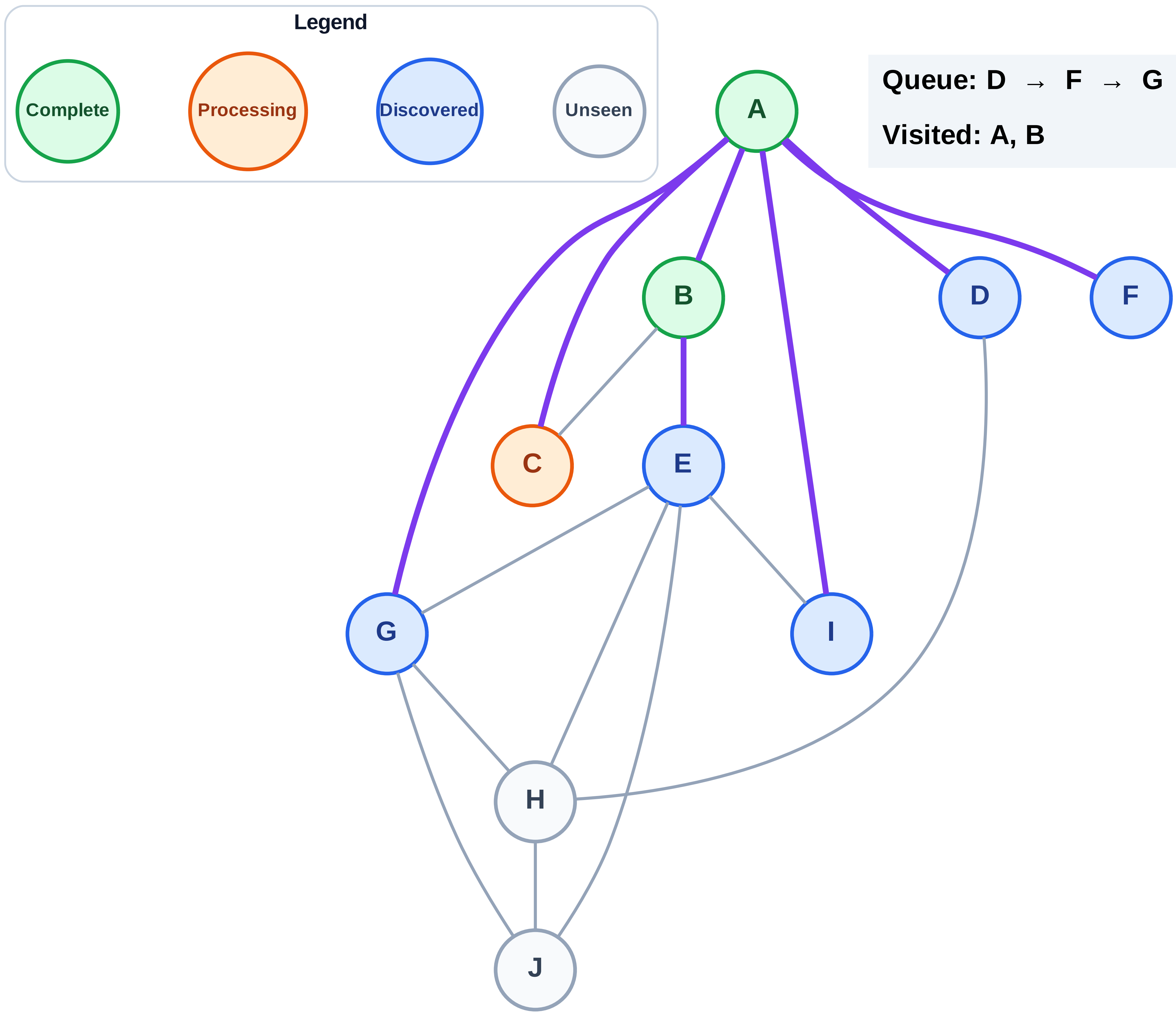
BFS Traversal

Step 5: Discover E from B



BFS Traversal

Step 6: Process node C

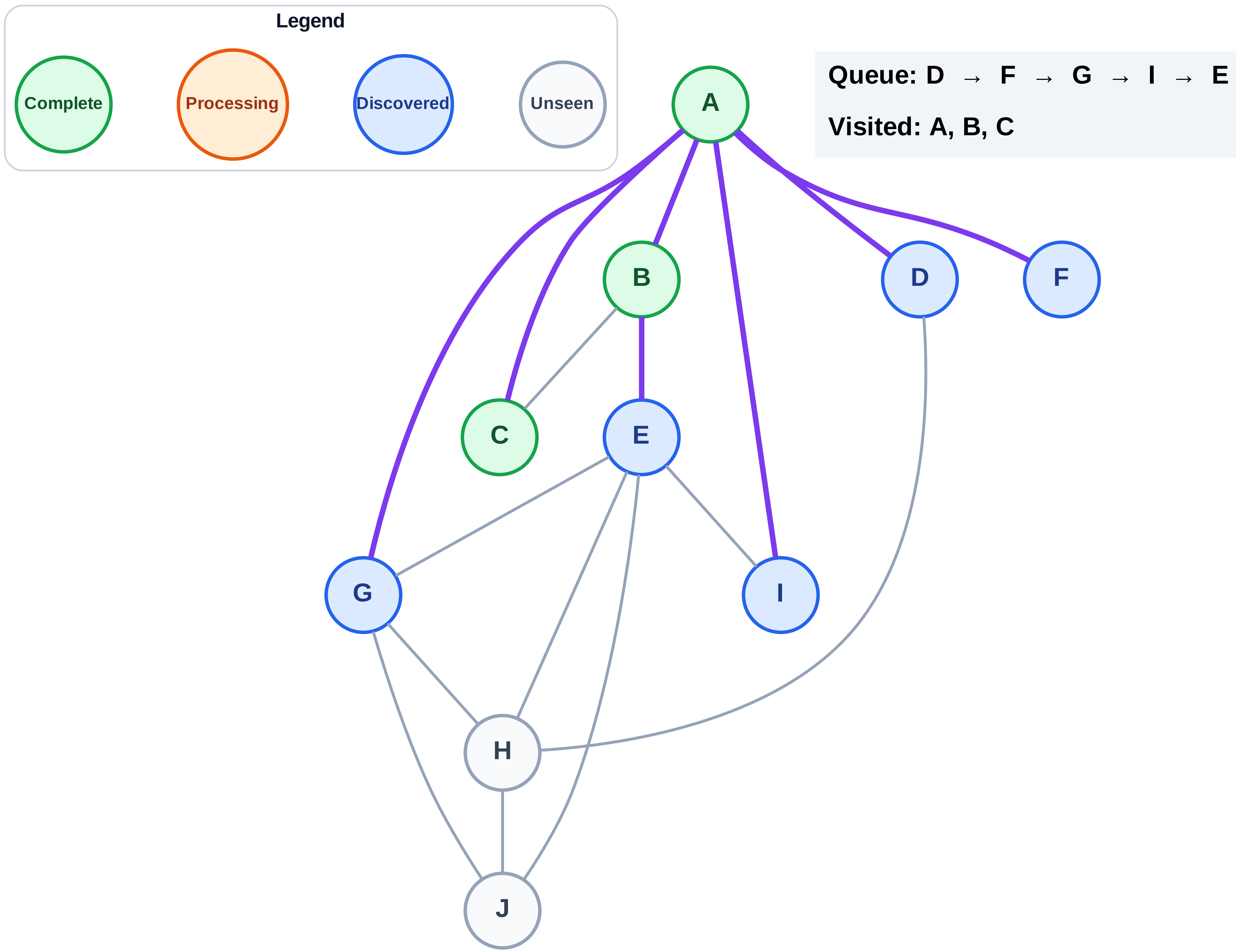


Queue: D → F → G → I → E

Visited: A, B

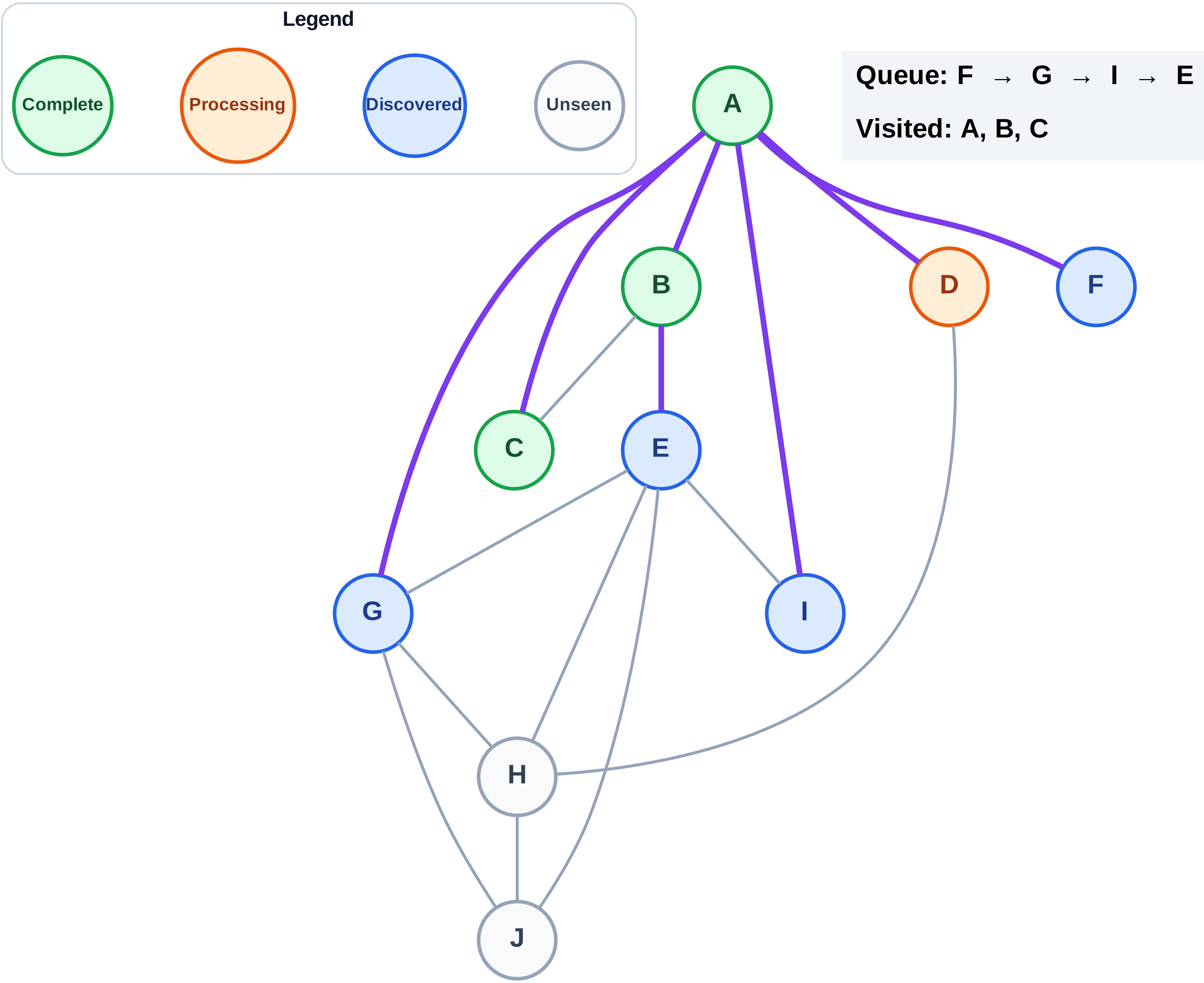
BFS Traversal

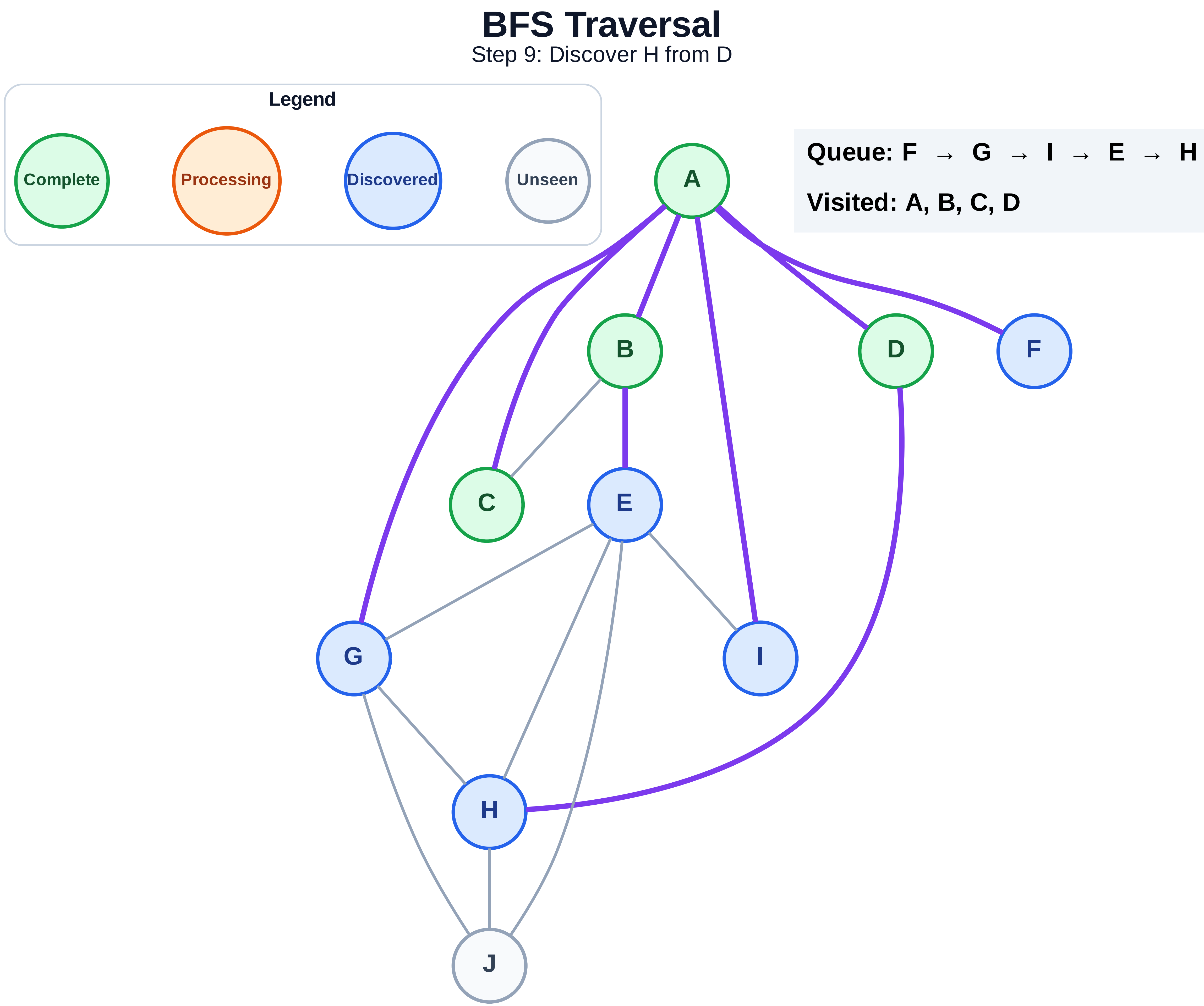
Step 7: No new neighbors from C

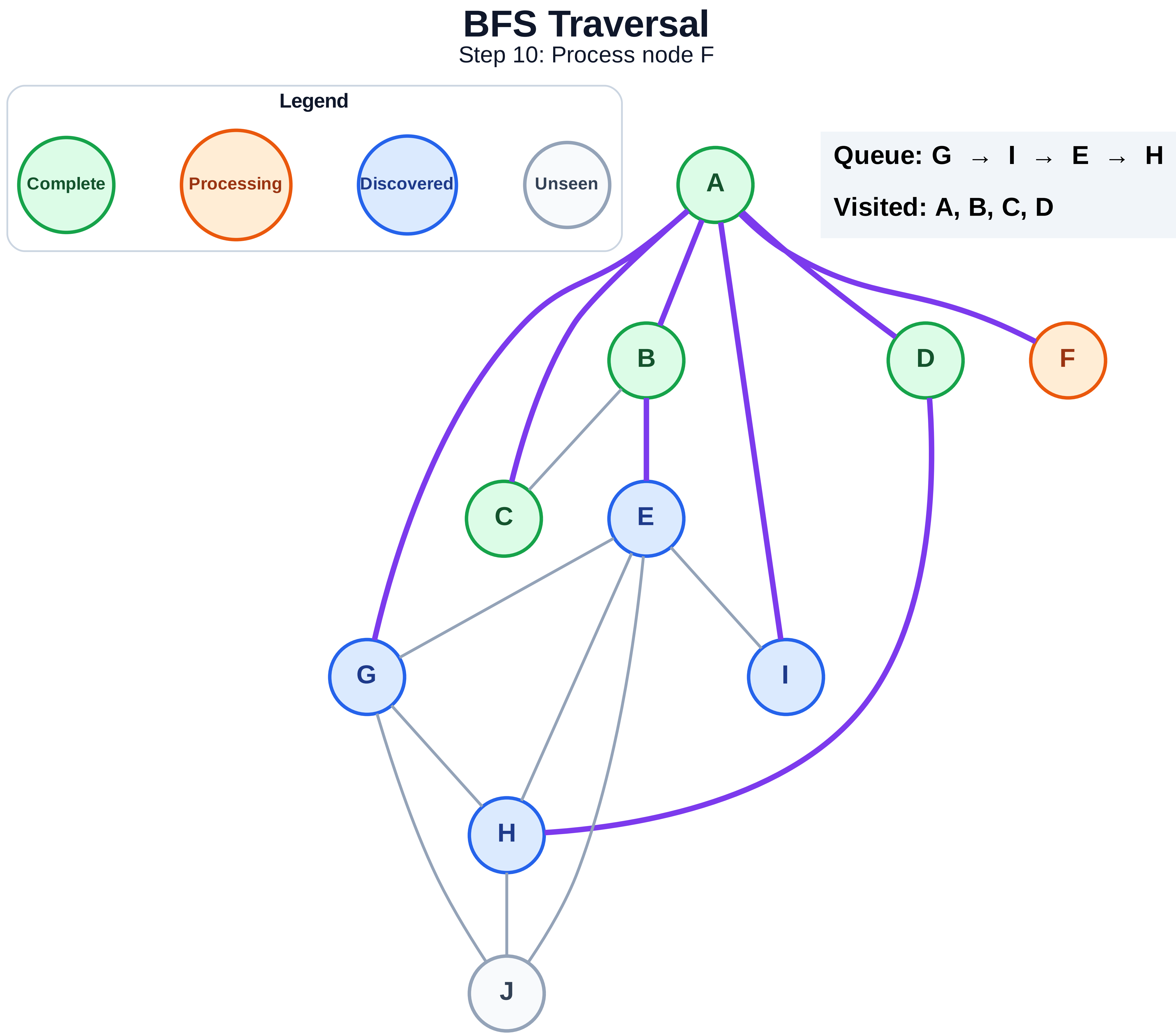


BFS Traversal

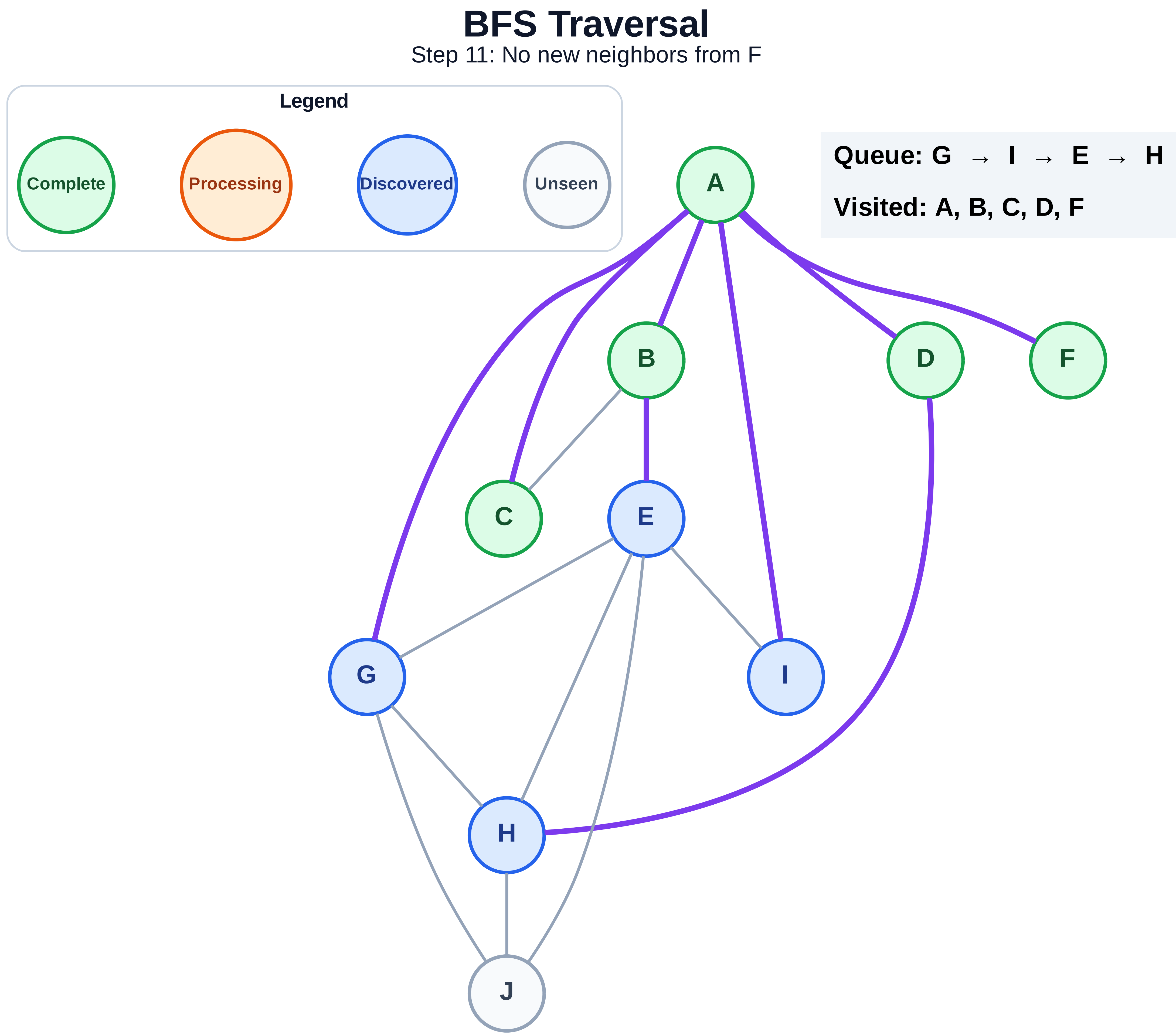
Step 8: Process node D

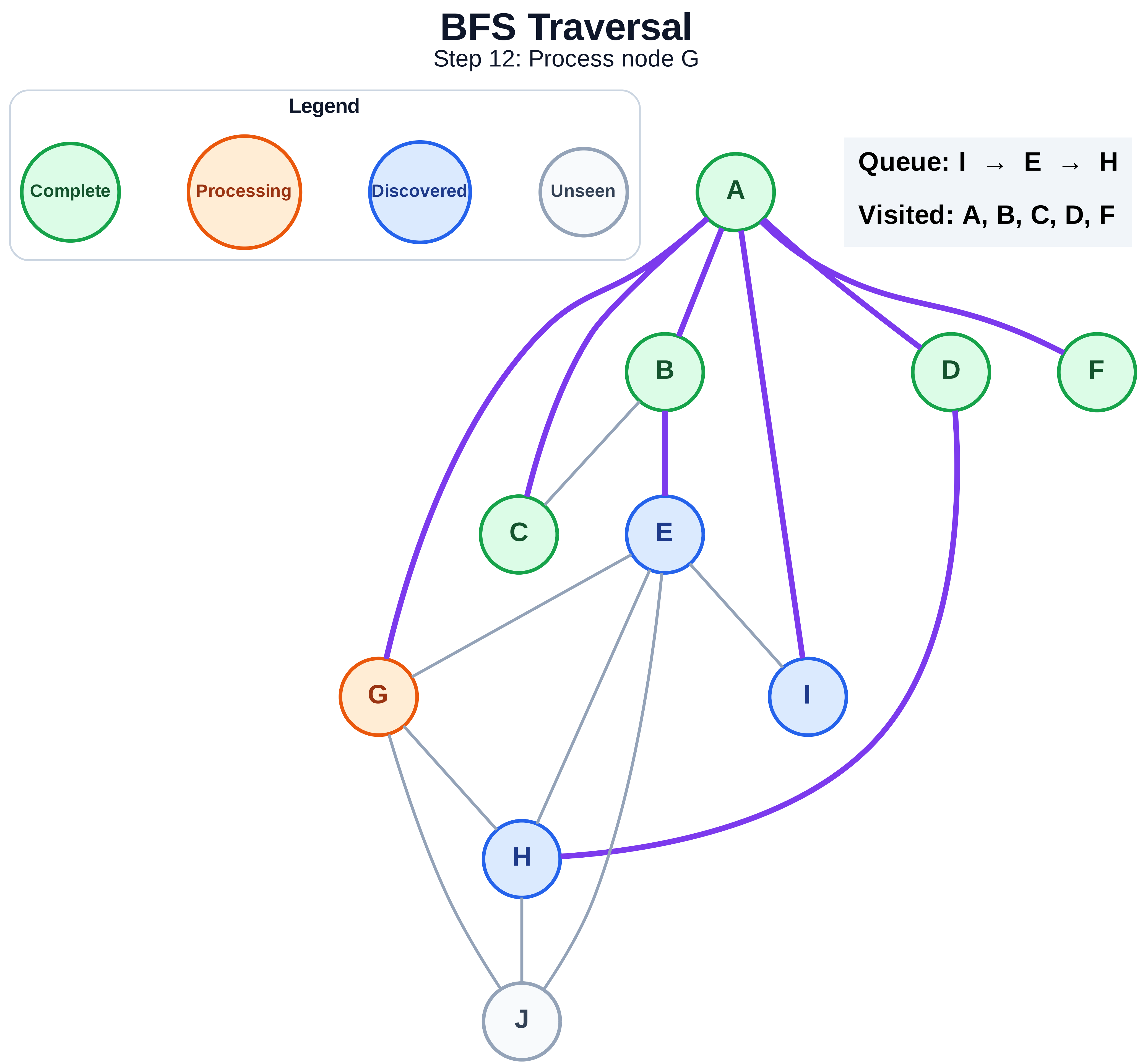



BFS Traversal
Step 9: Discover H from D
Legend
Complete
Processing
Discovered
Unseen
Queue: F → G → I → E → H
Visited: A, B, C, D
A
B
C
D
E
F
G
H
I
J



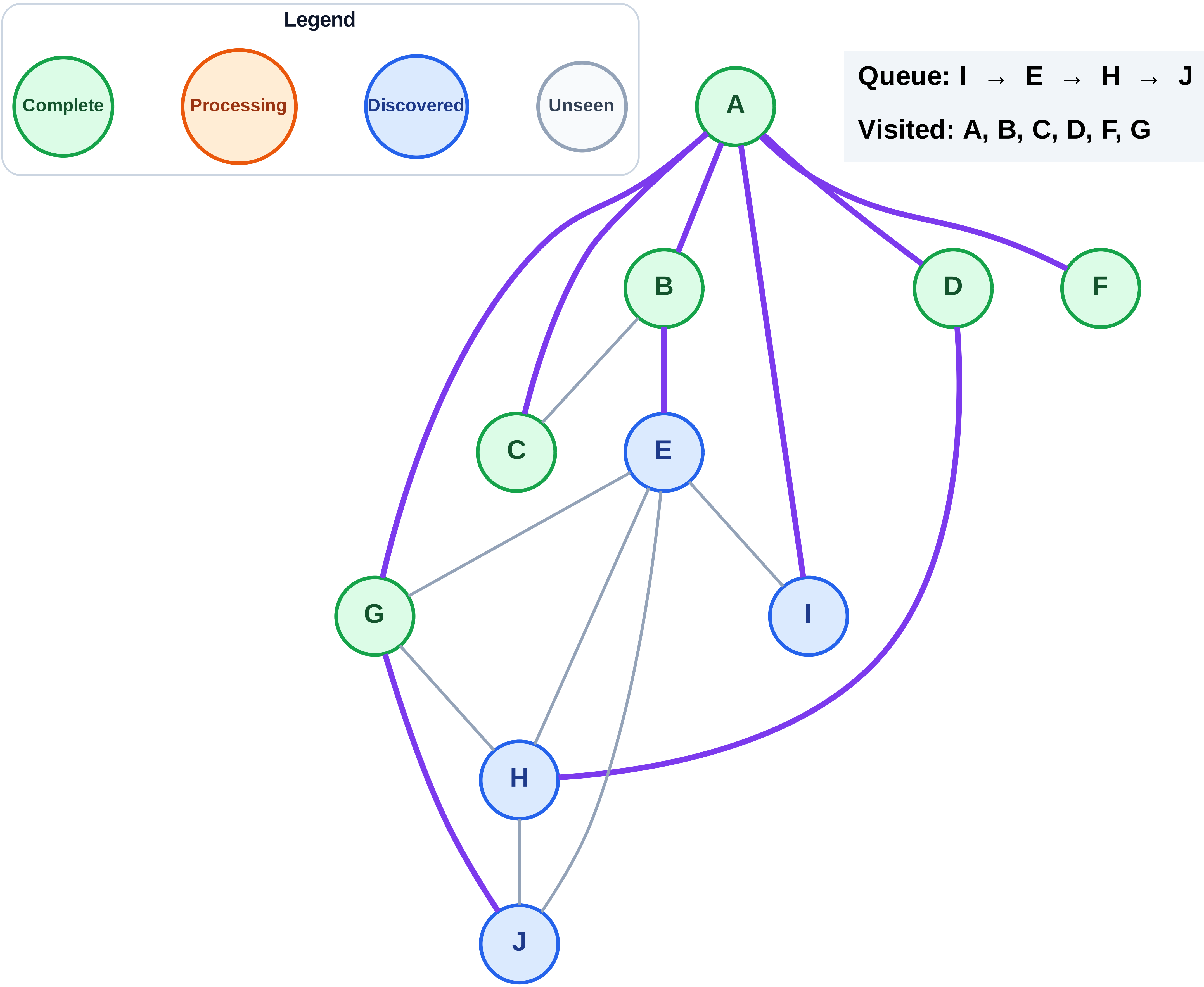
BFS Traversal
Step 10: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → I → E → H
Visited: A, B, C, D
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 11: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Queue: G → I → E → H
Visited: A, B, C, D, F
A
B
D
F
C
E
G
I
H
J


BFS Traversal
Step 12: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: I → E → H
Visited: A, B, C, D, F
A
B
D
F
C
E
G
I
H
J

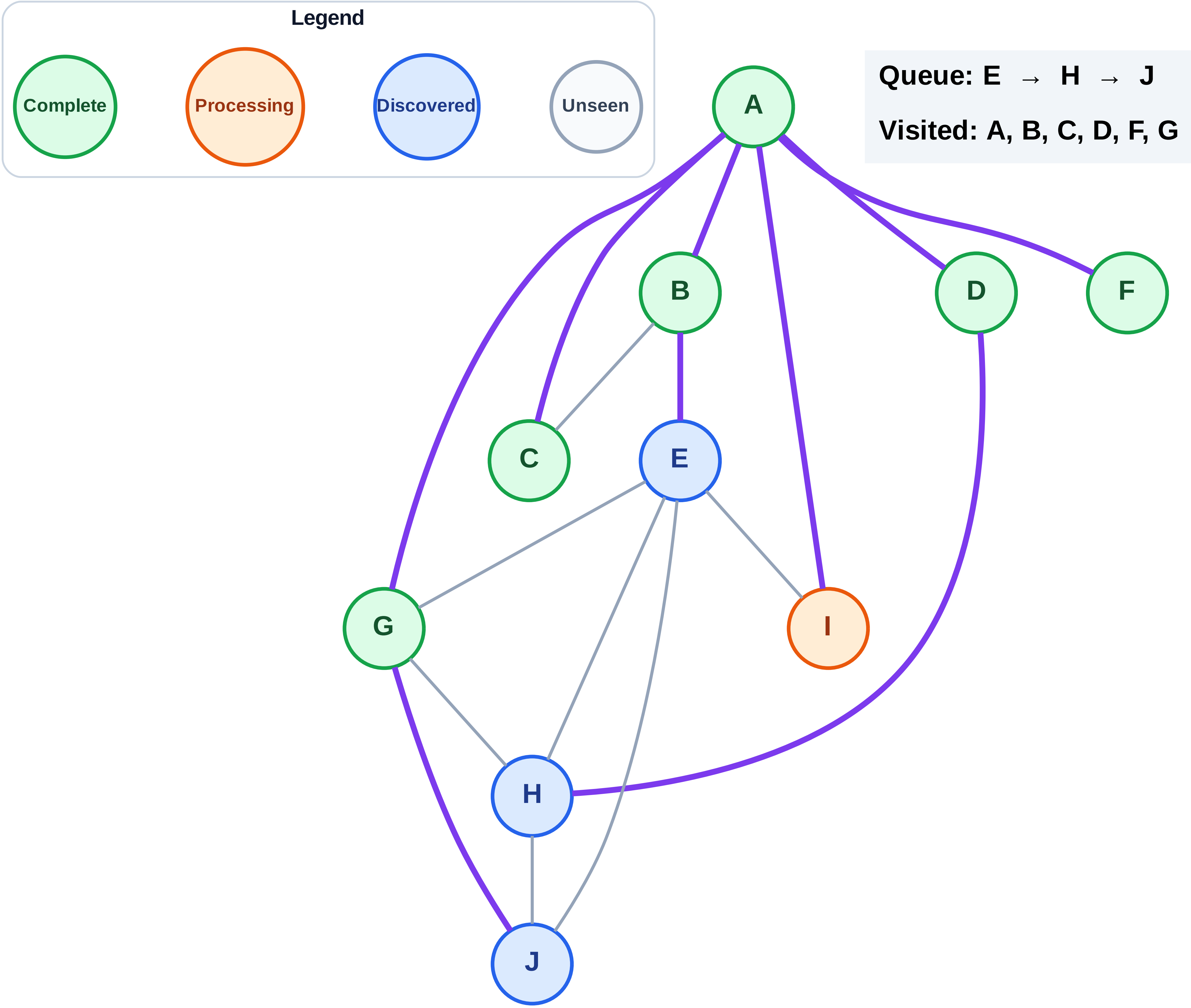
BFS Traversal

Step 13: Discover J from G



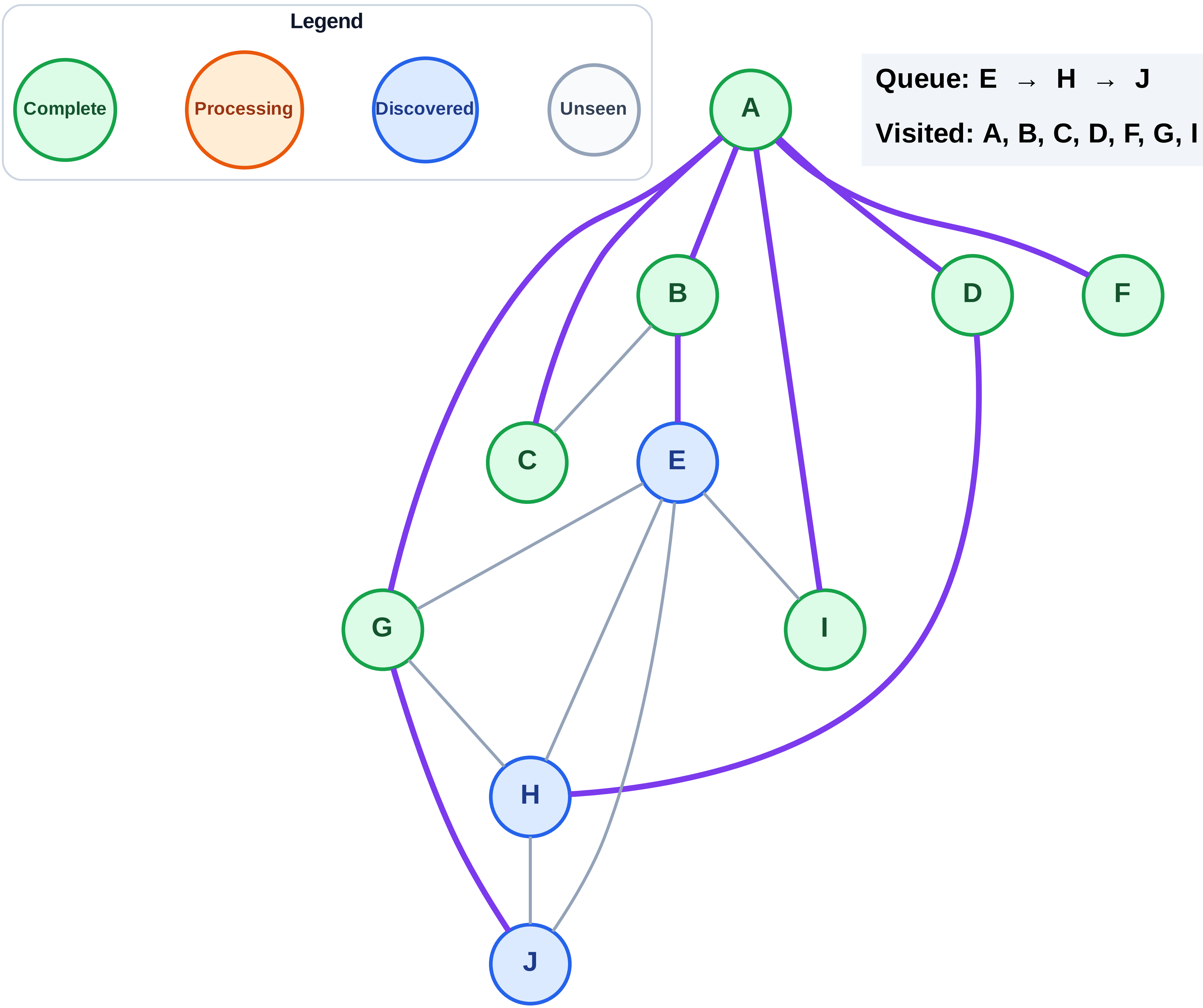
BFS Traversal

Step 14: Process node I



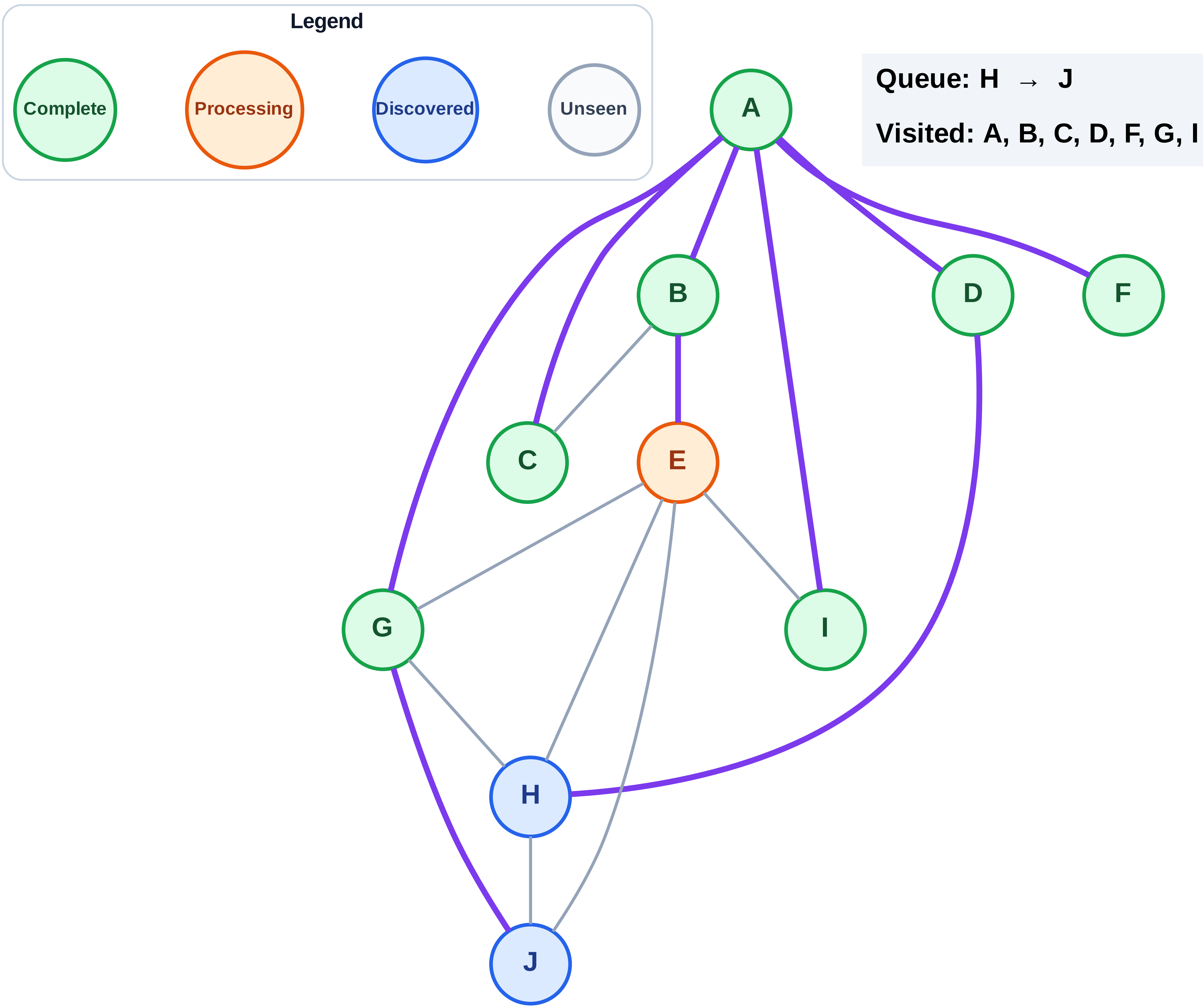
BFS Traversal

Step 15: No new neighbors from I



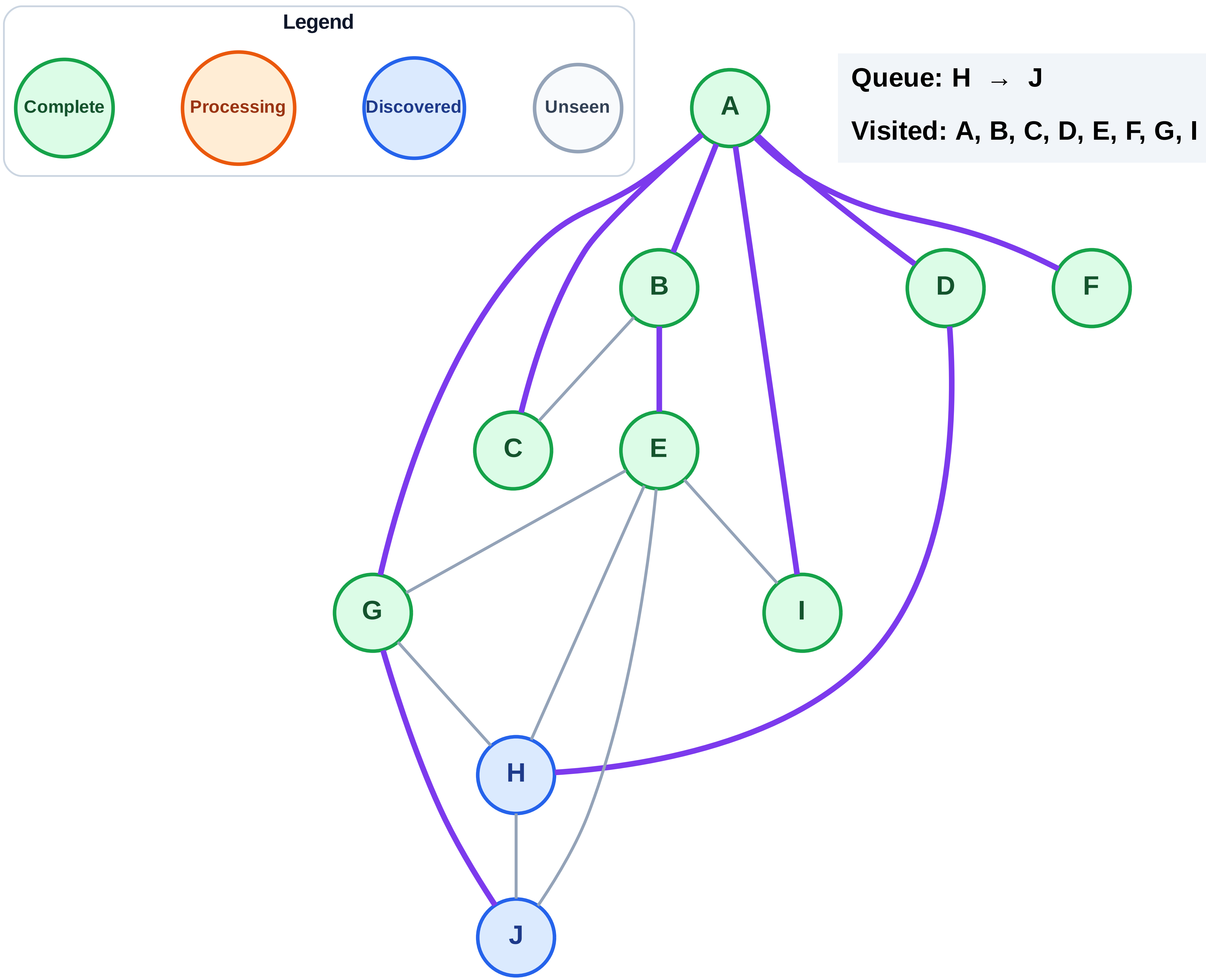
BFS Traversal

Step 16: Process node E



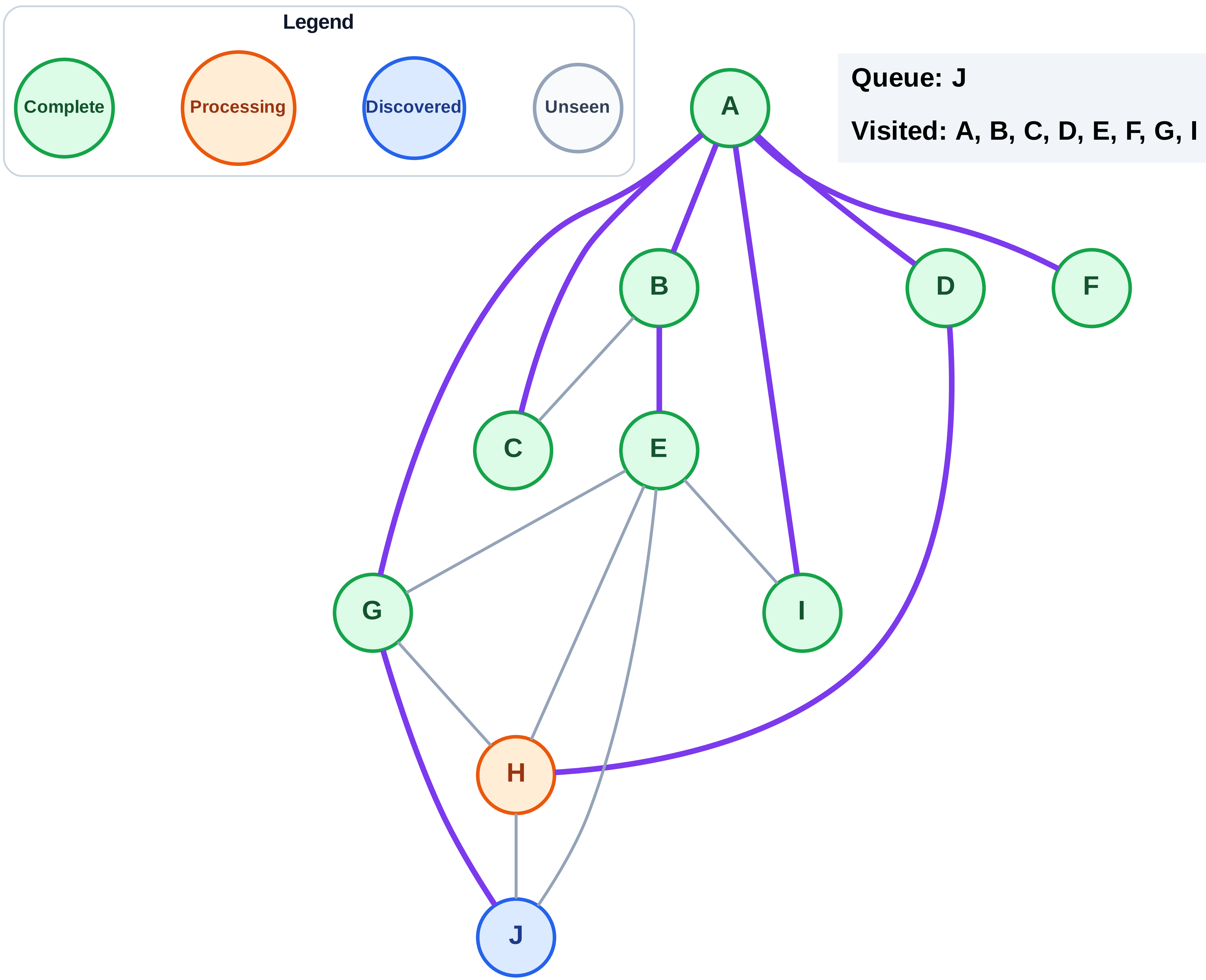
BFS Traversal

Step 17: No new neighbors from E



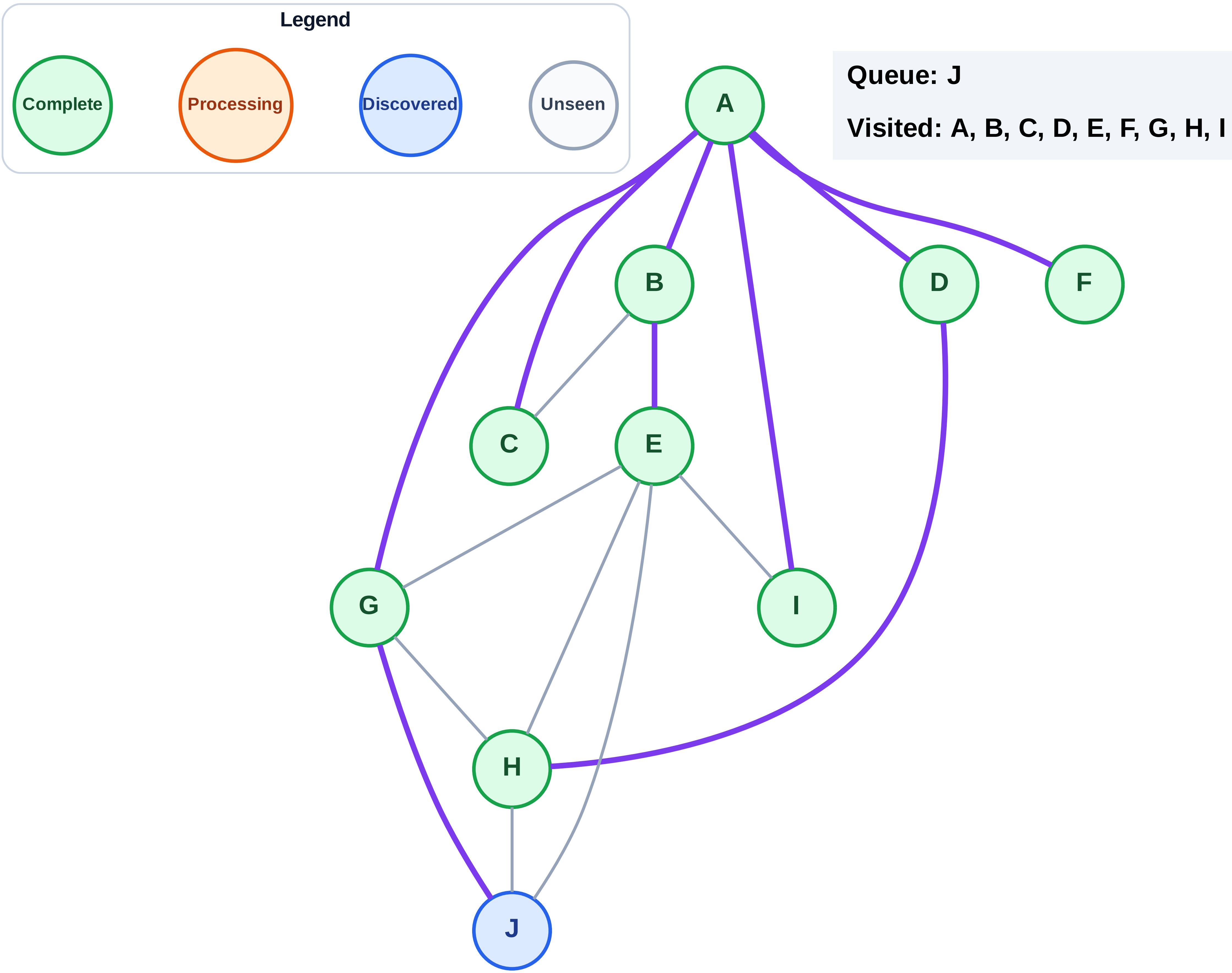
BFS Traversal

Step 18: Process node H



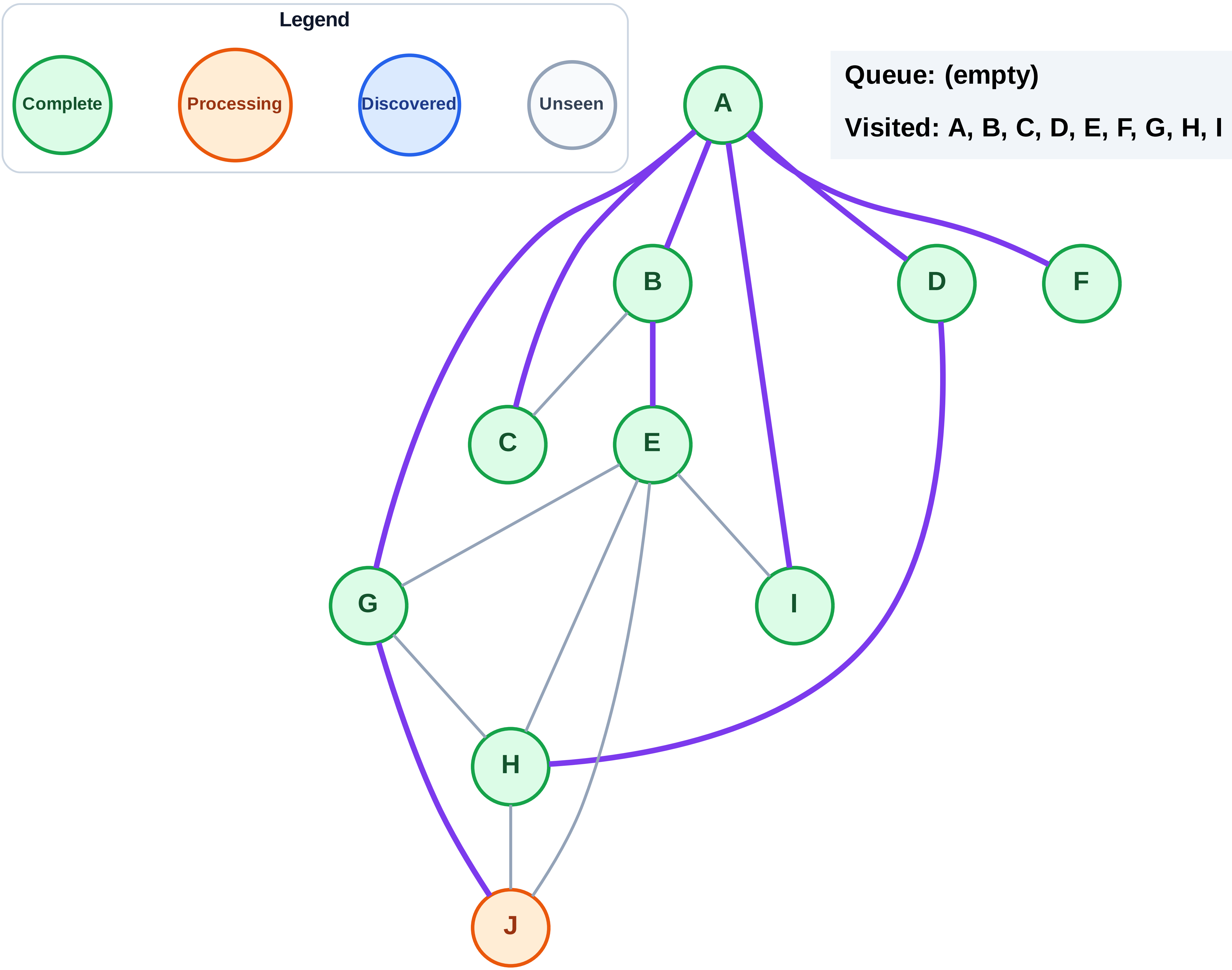
BFS Traversal

Step 19: No new neighbors from H



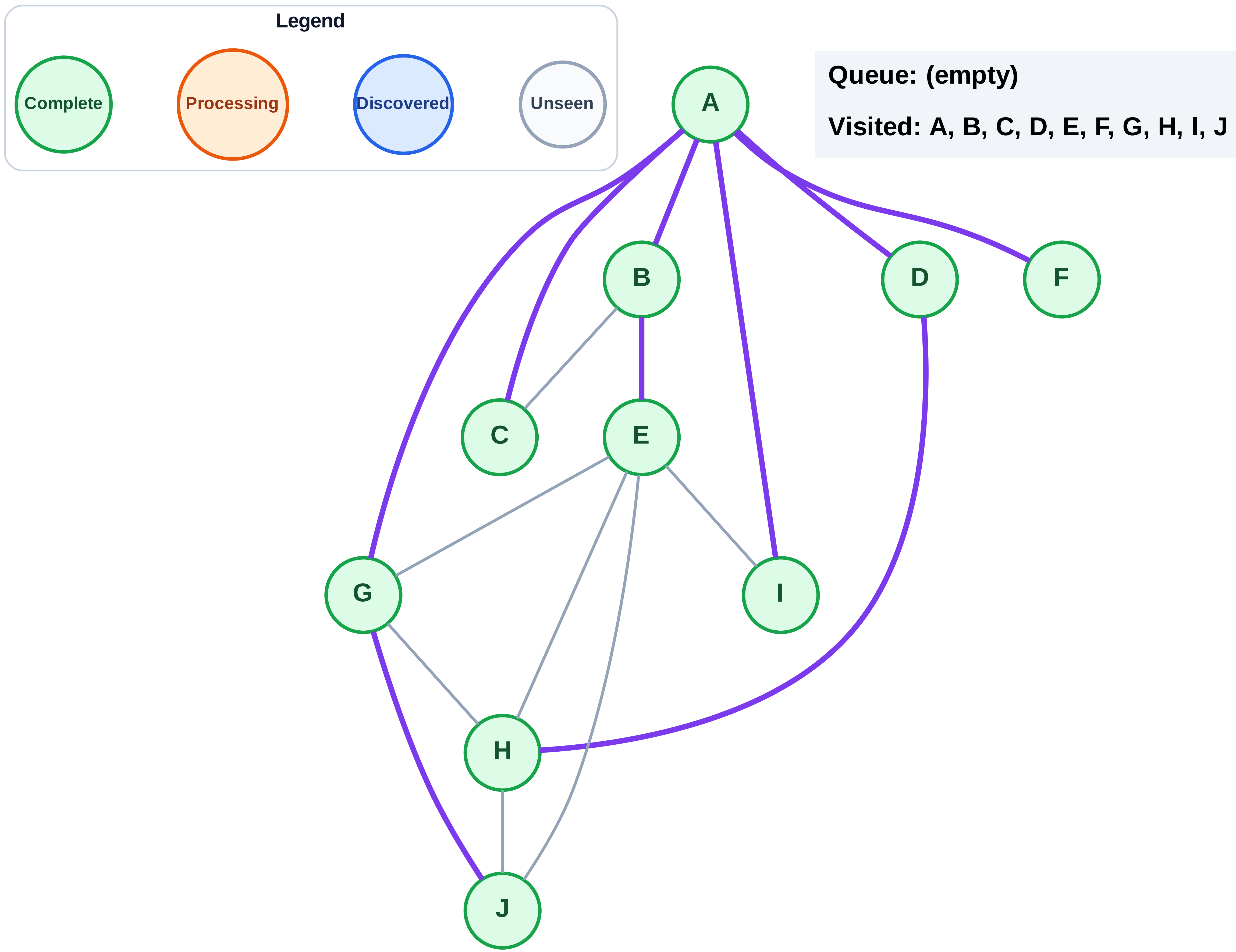
BFS Traversal

Step 20: Process node J



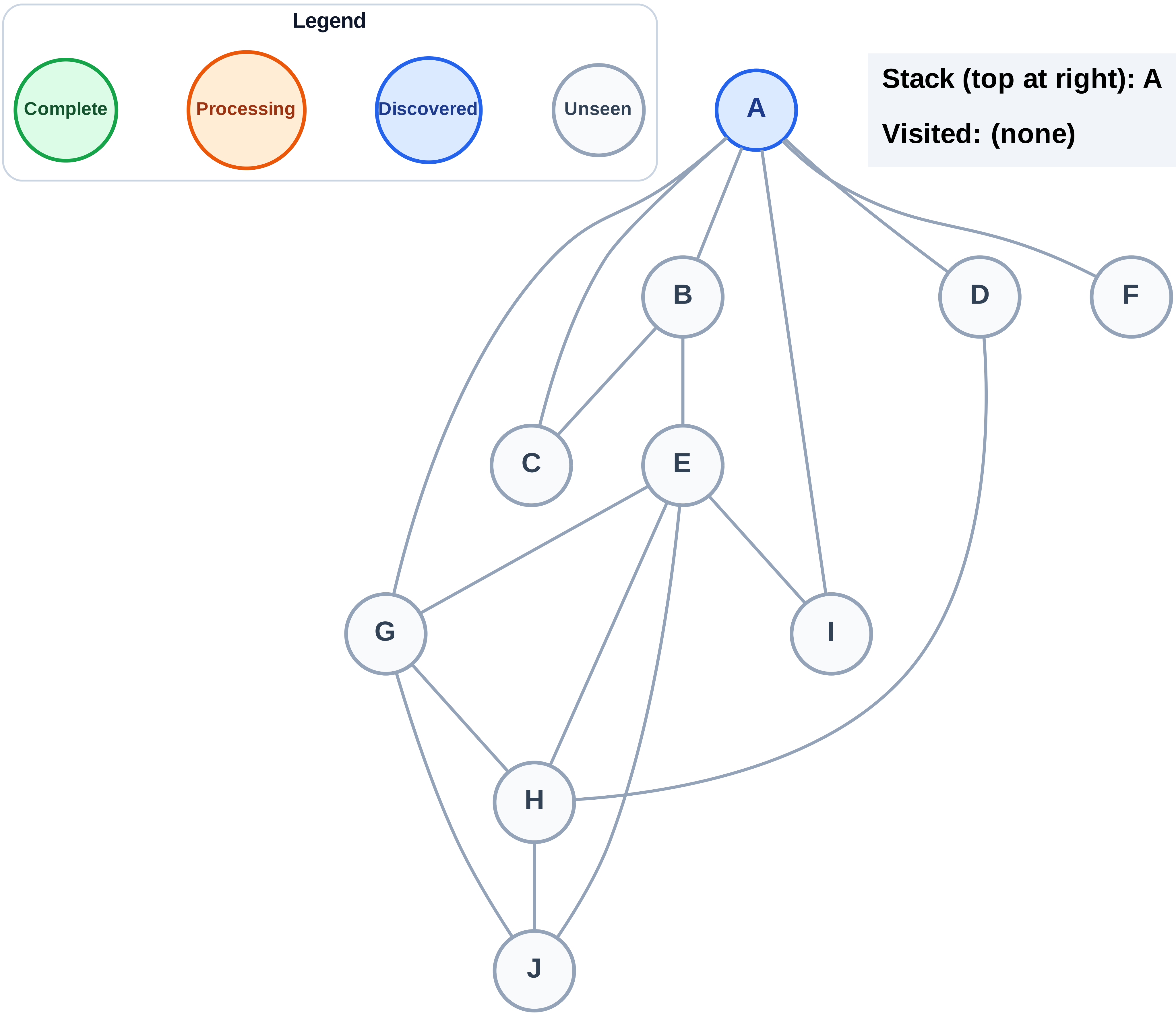
BFS Traversal

Step 21: No new neighbors from J



DFS Traversal

Step 1: Start at A

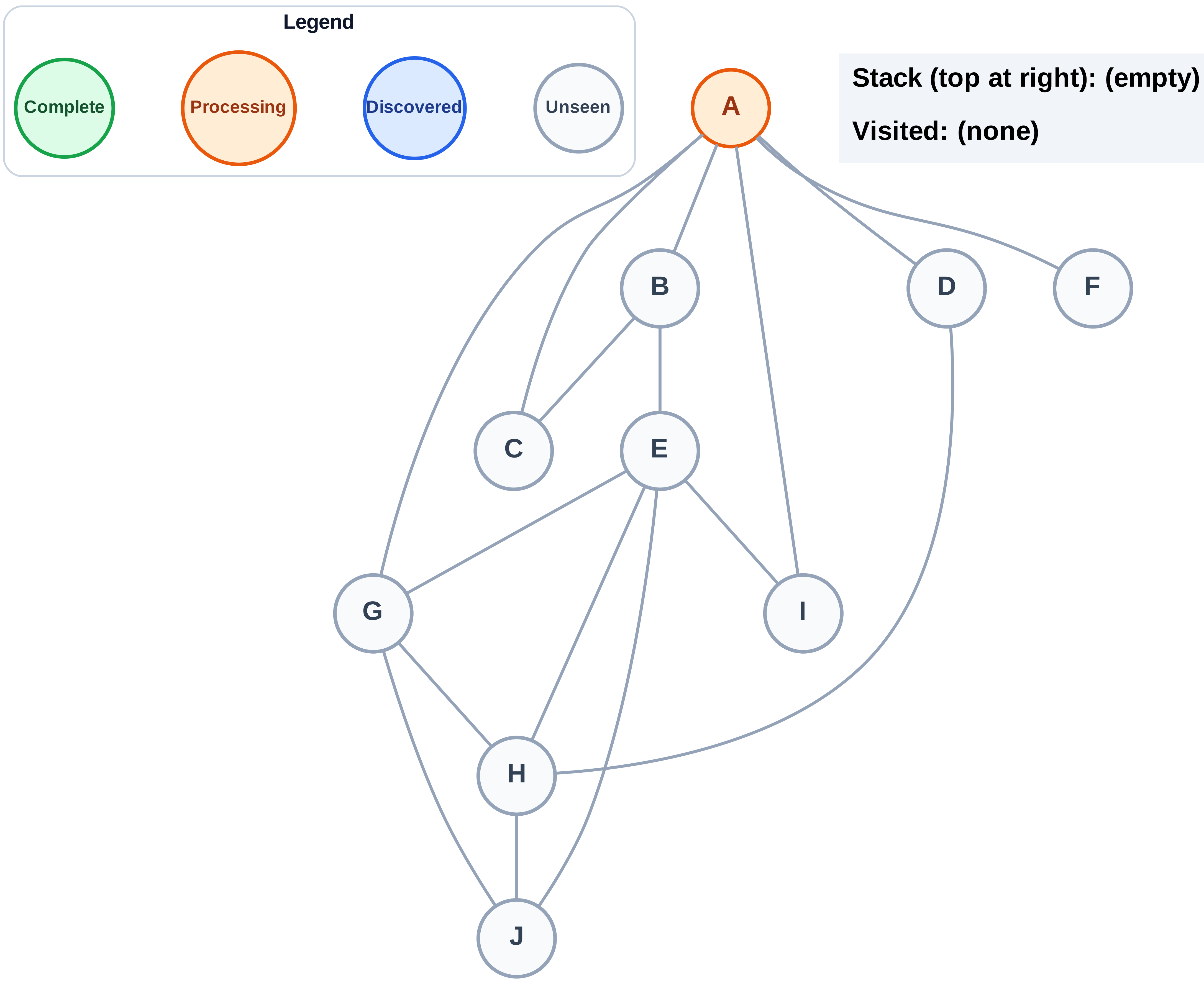


Stack (top at right): A

Visited: (none)

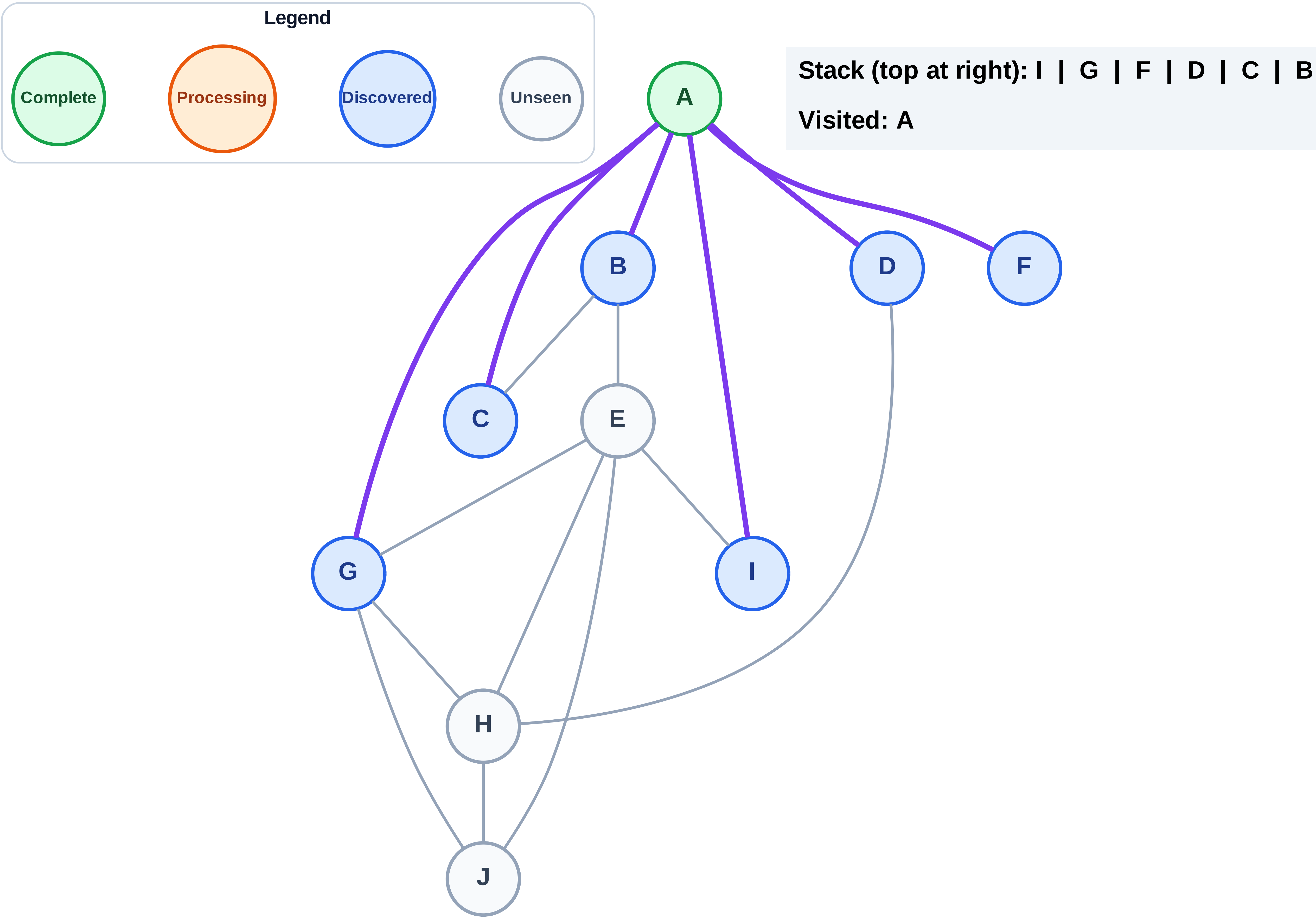
DFS Traversal

Step 2: Process node A



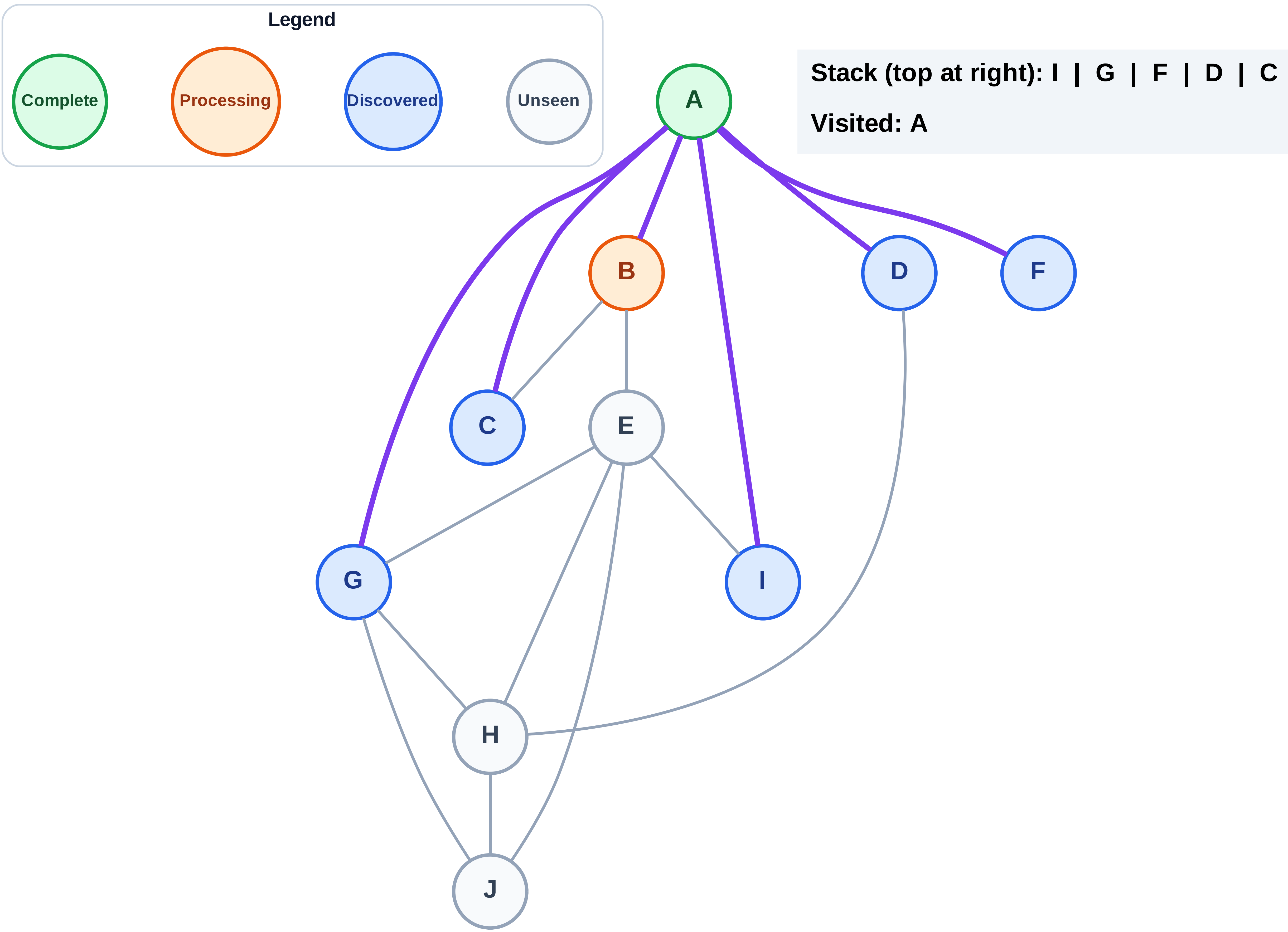
DFS Traversal

Step 3: Discover I, G, F, D, C, B from A



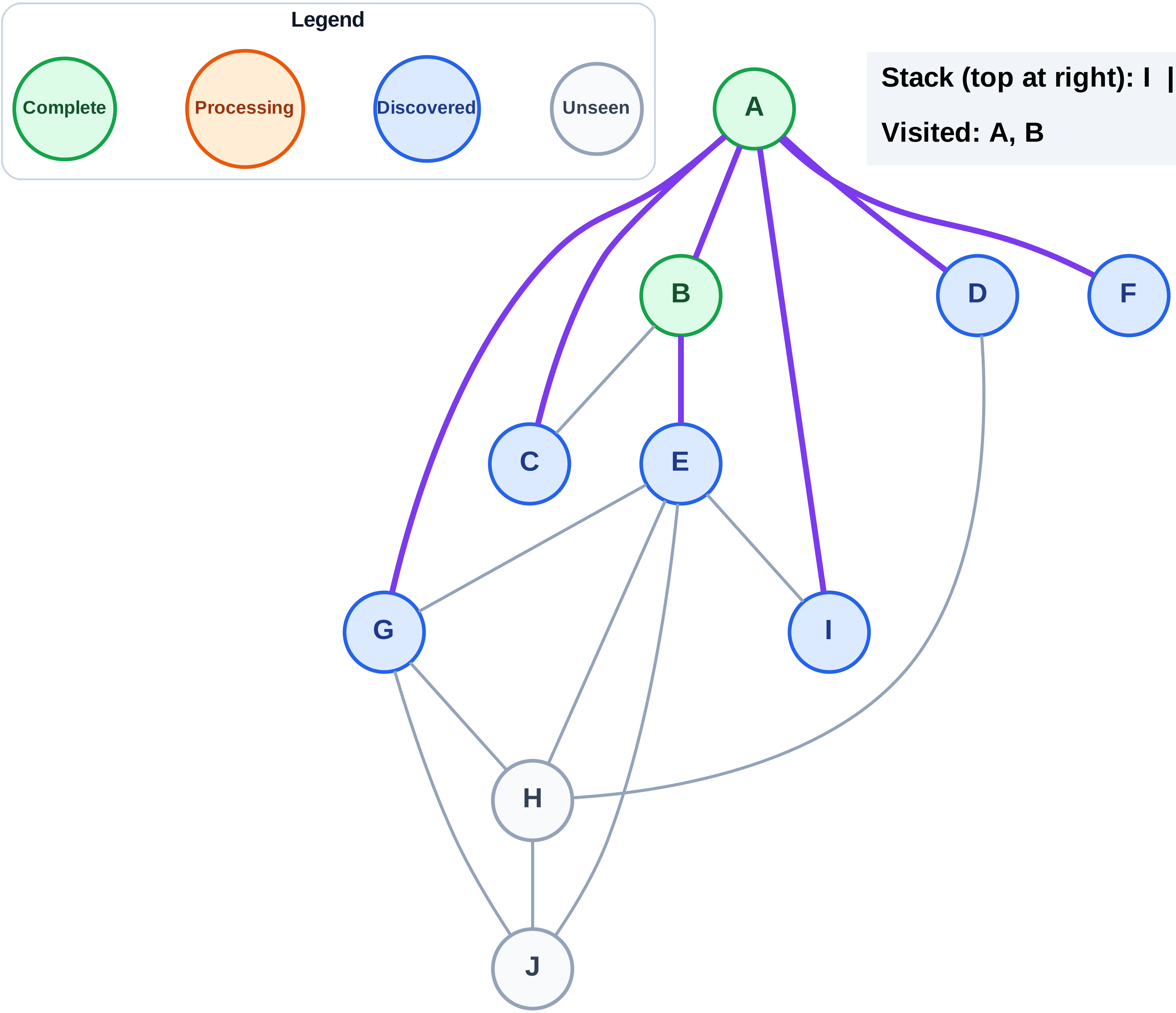
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover E from B

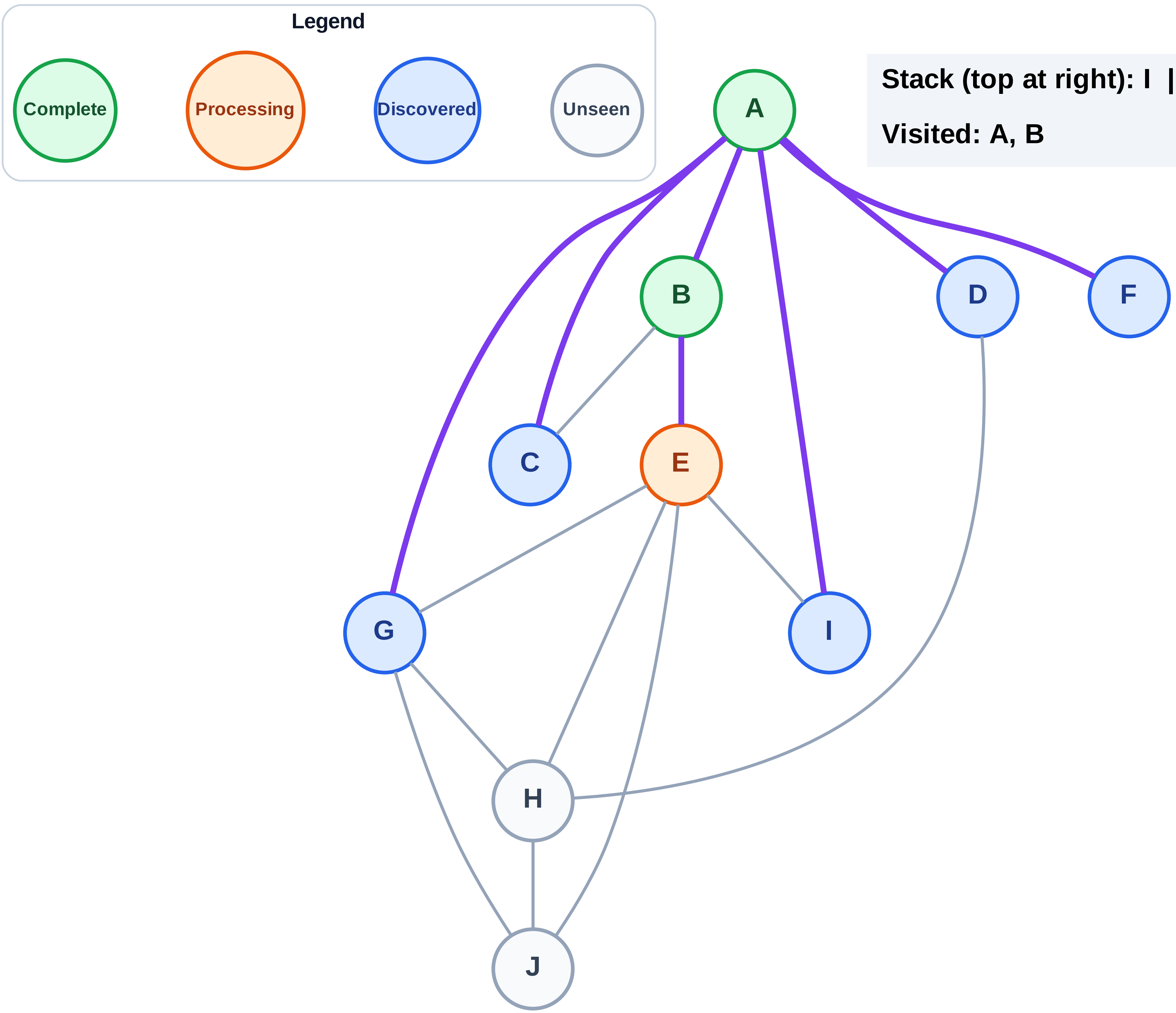


Stack (top at right): I | G | F | D | C | E

Visited: A, B

DFS Traversal

Step 6: Process node E

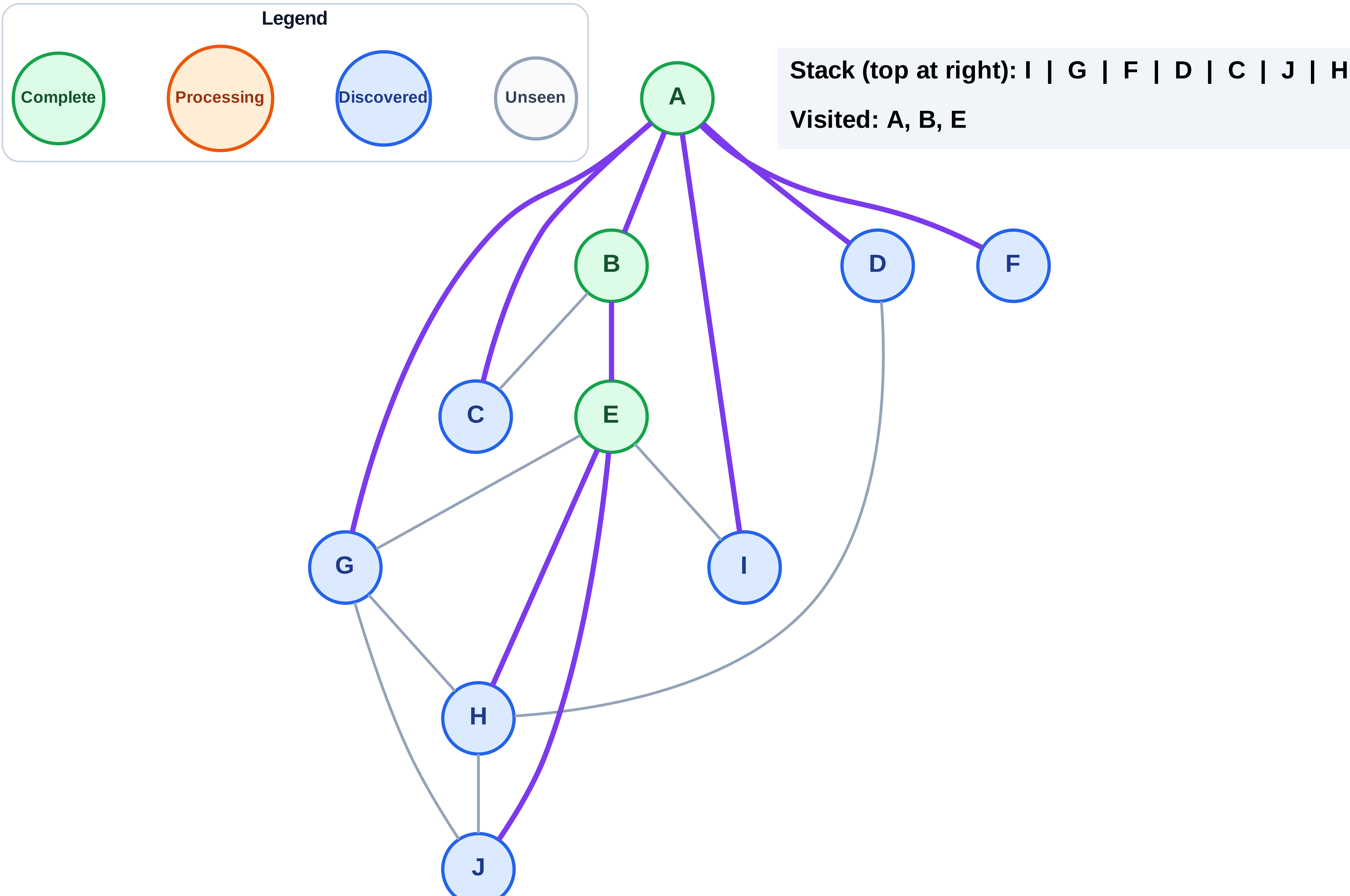


Stack (top at right): I | G | F | D | C

Visited: A, B

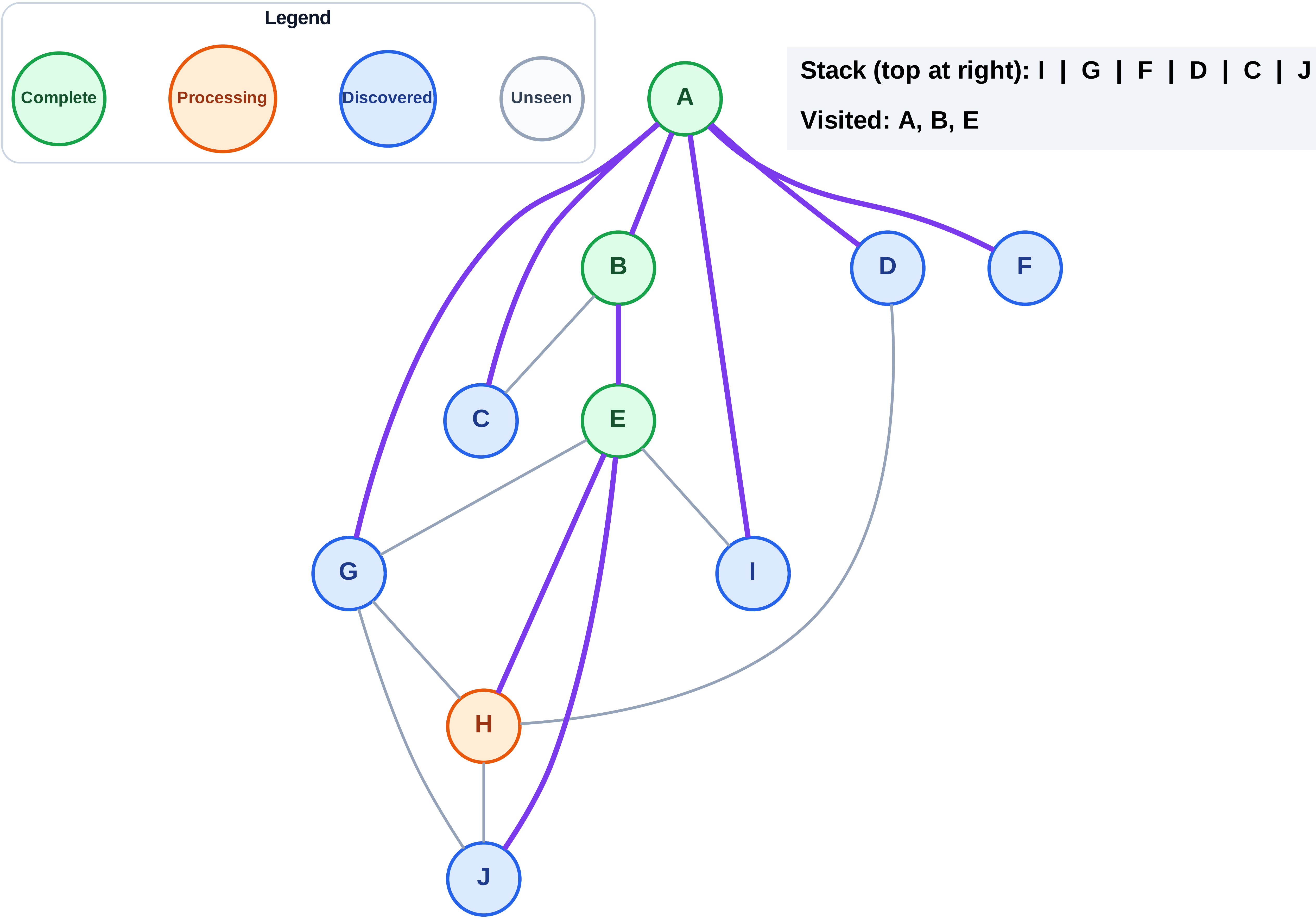
DFS Traversal

Step 7: Discover J, H from E



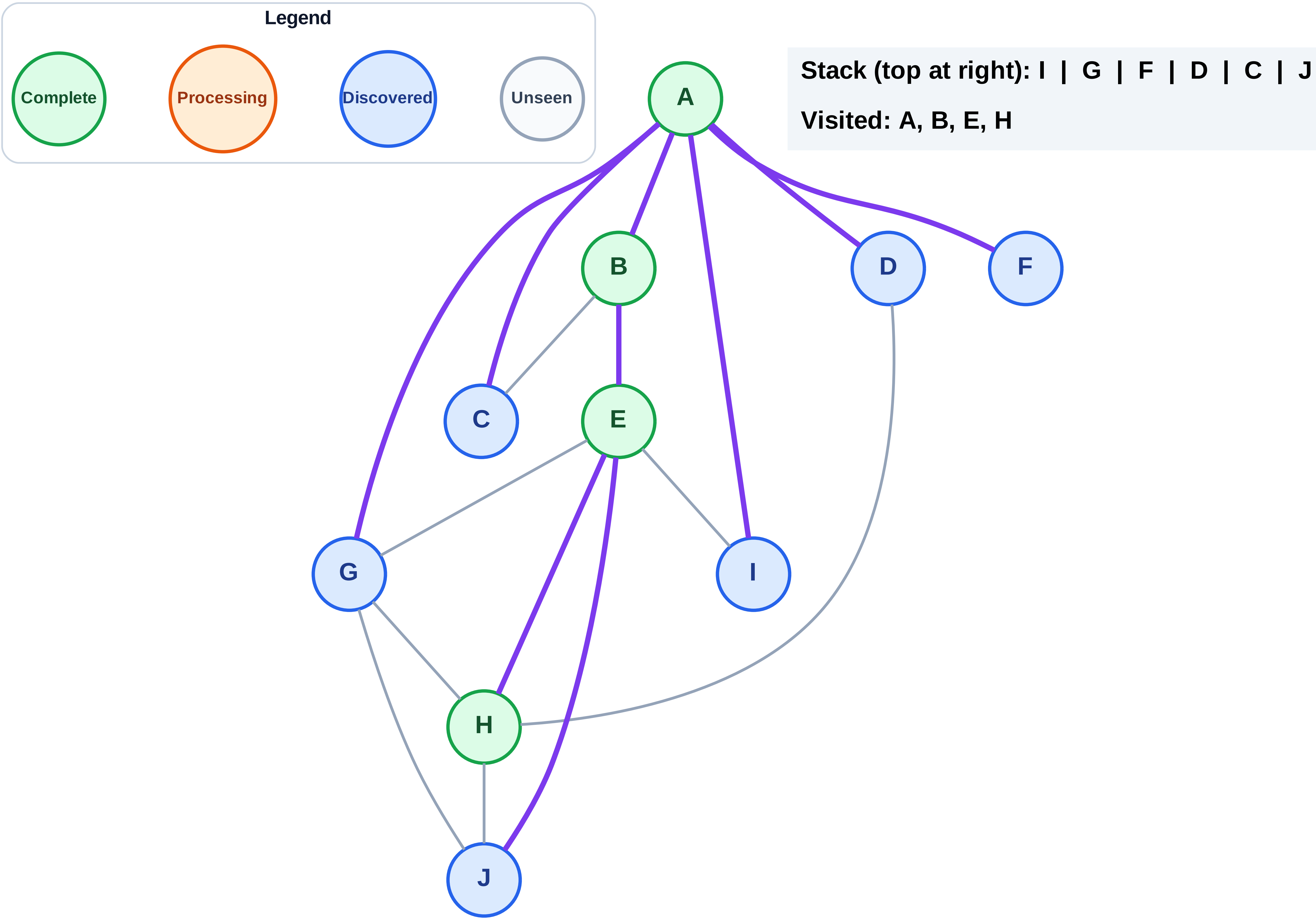
DFS Traversal

Step 8: Process node H



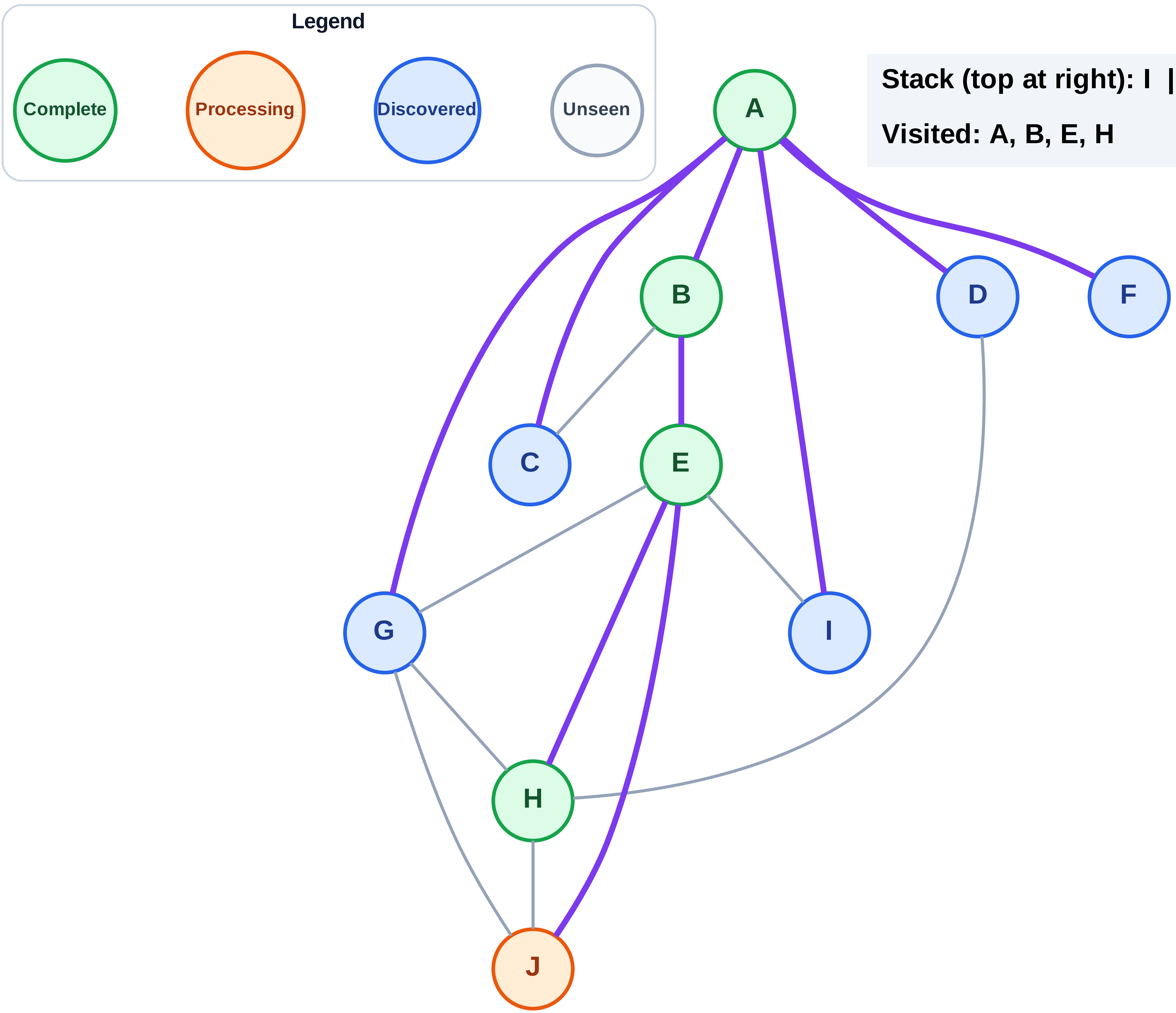
DFS Traversal

Step 9: No new neighbors from H



DFS Traversal

Step 10: Process node J

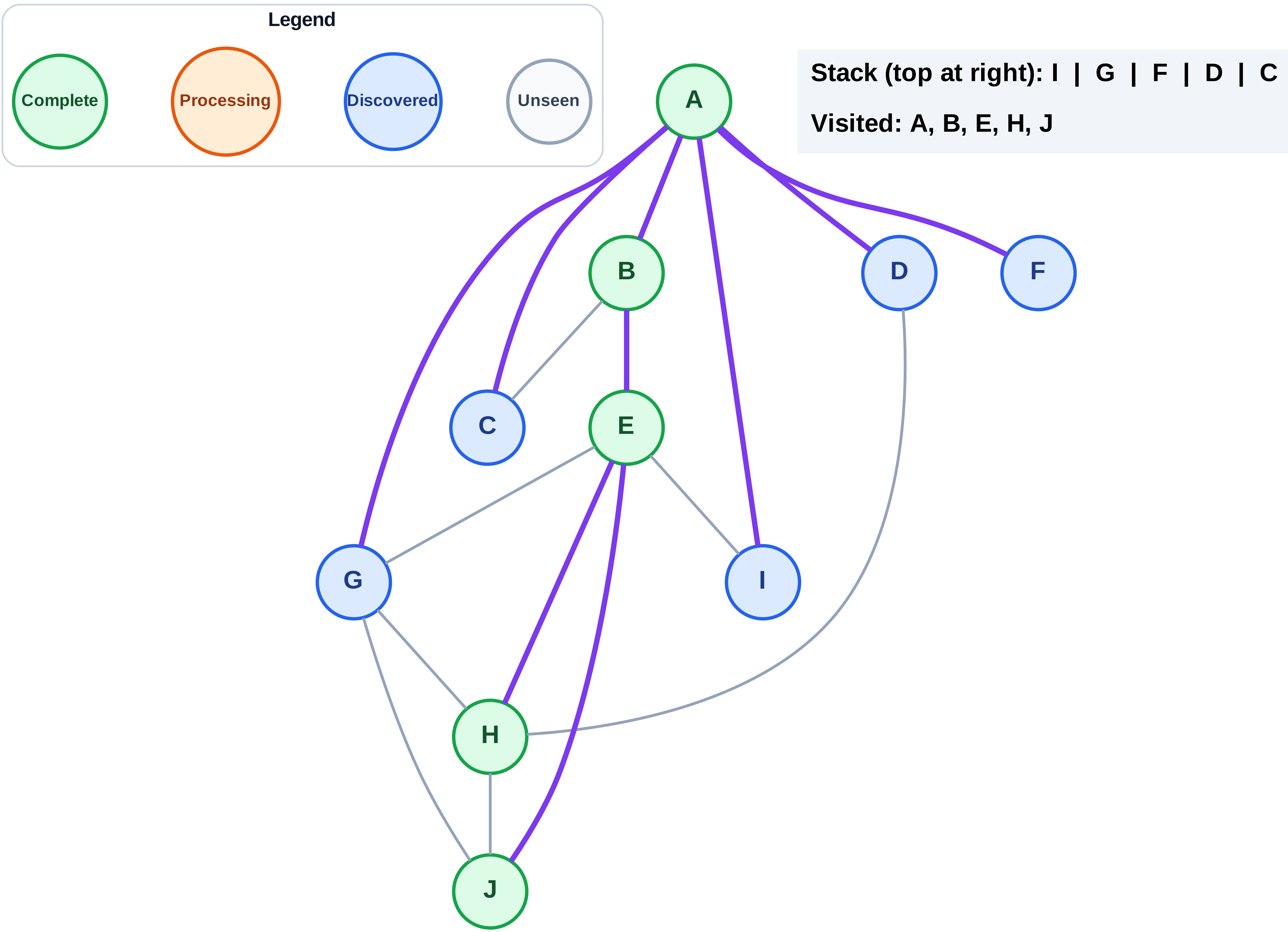


Stack (top at right): I | G | F | D | C

Visited: A, B, E, H

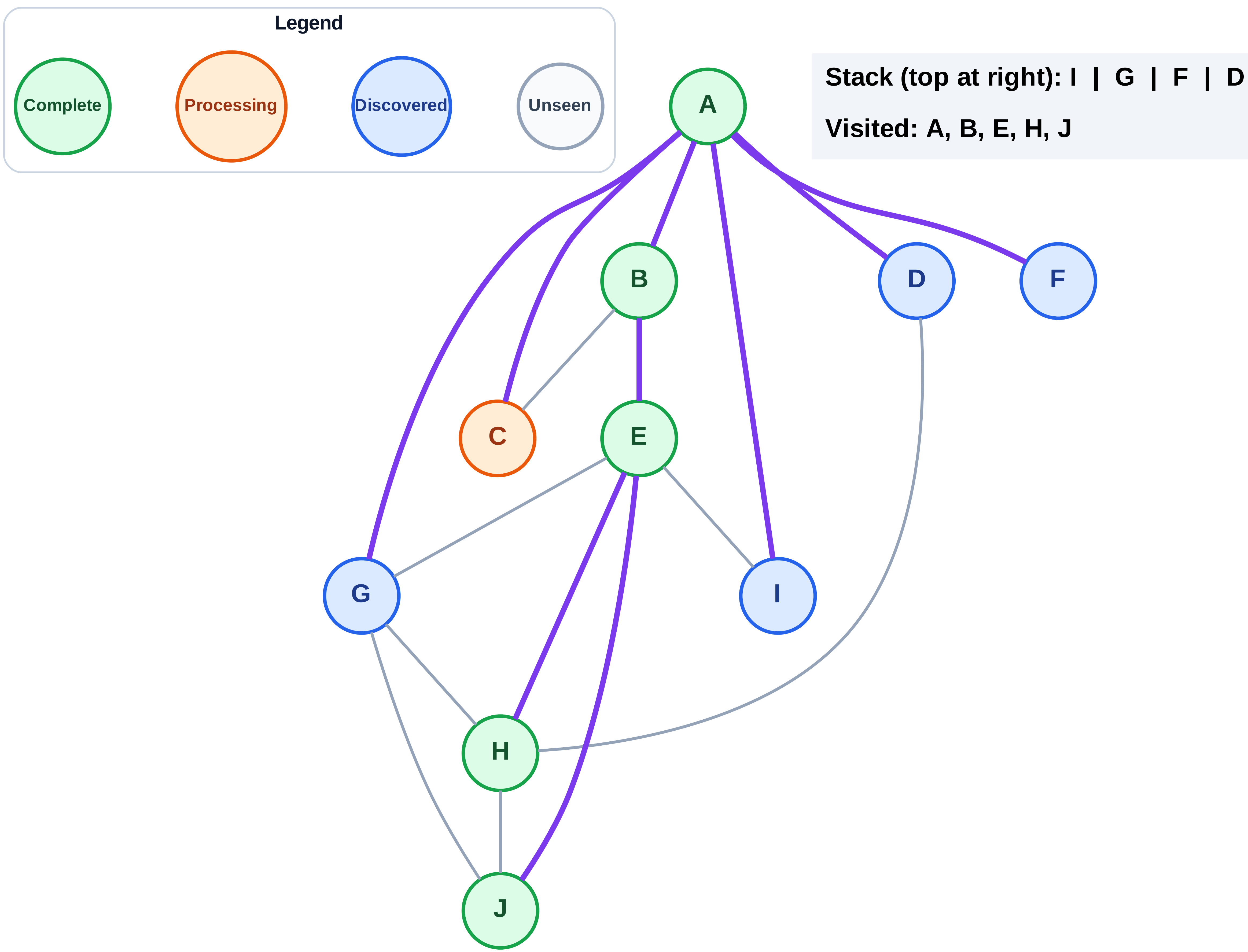
DFS Traversal

Step 11: No new neighbors from J



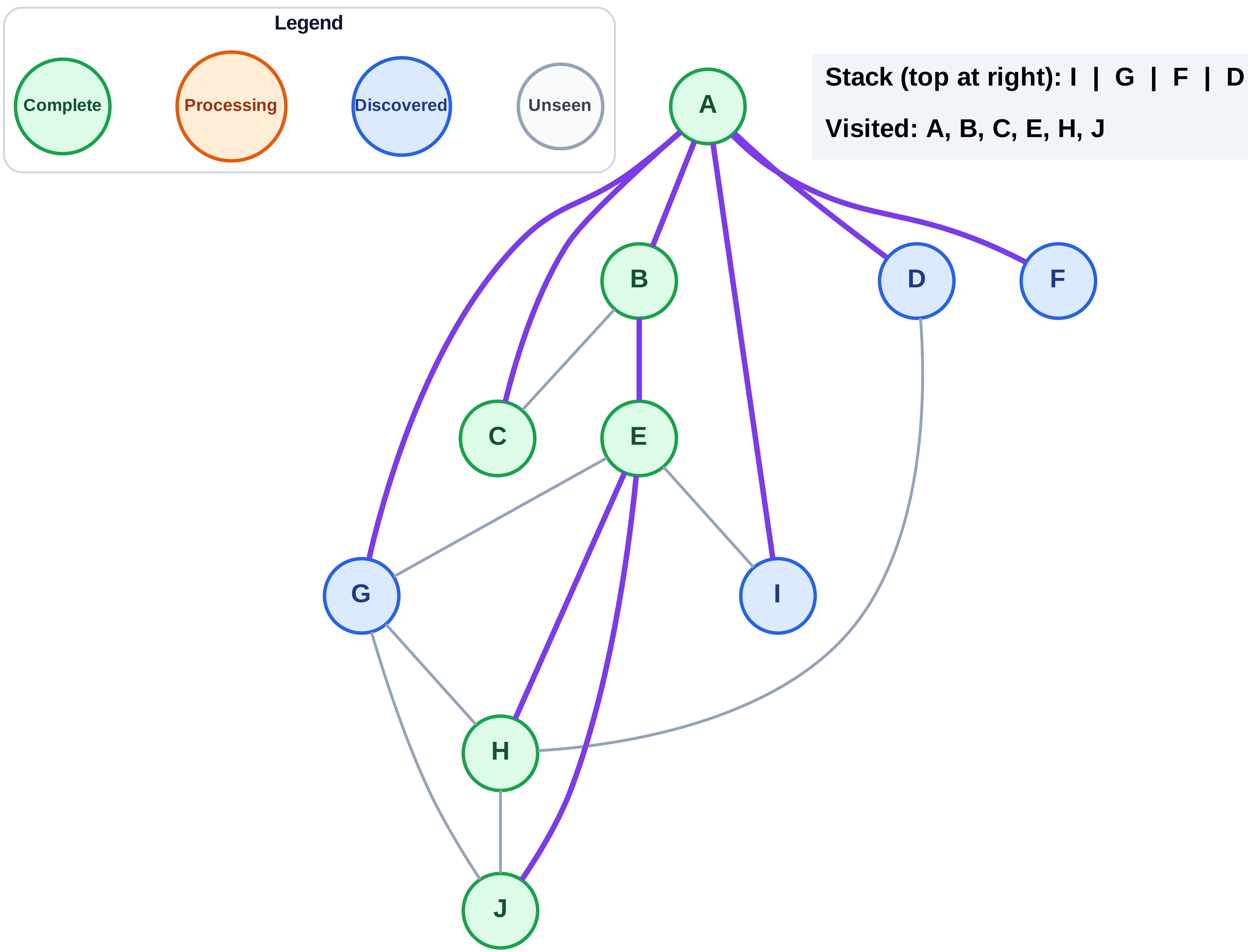
DFS Traversal

Step 12: Process node C



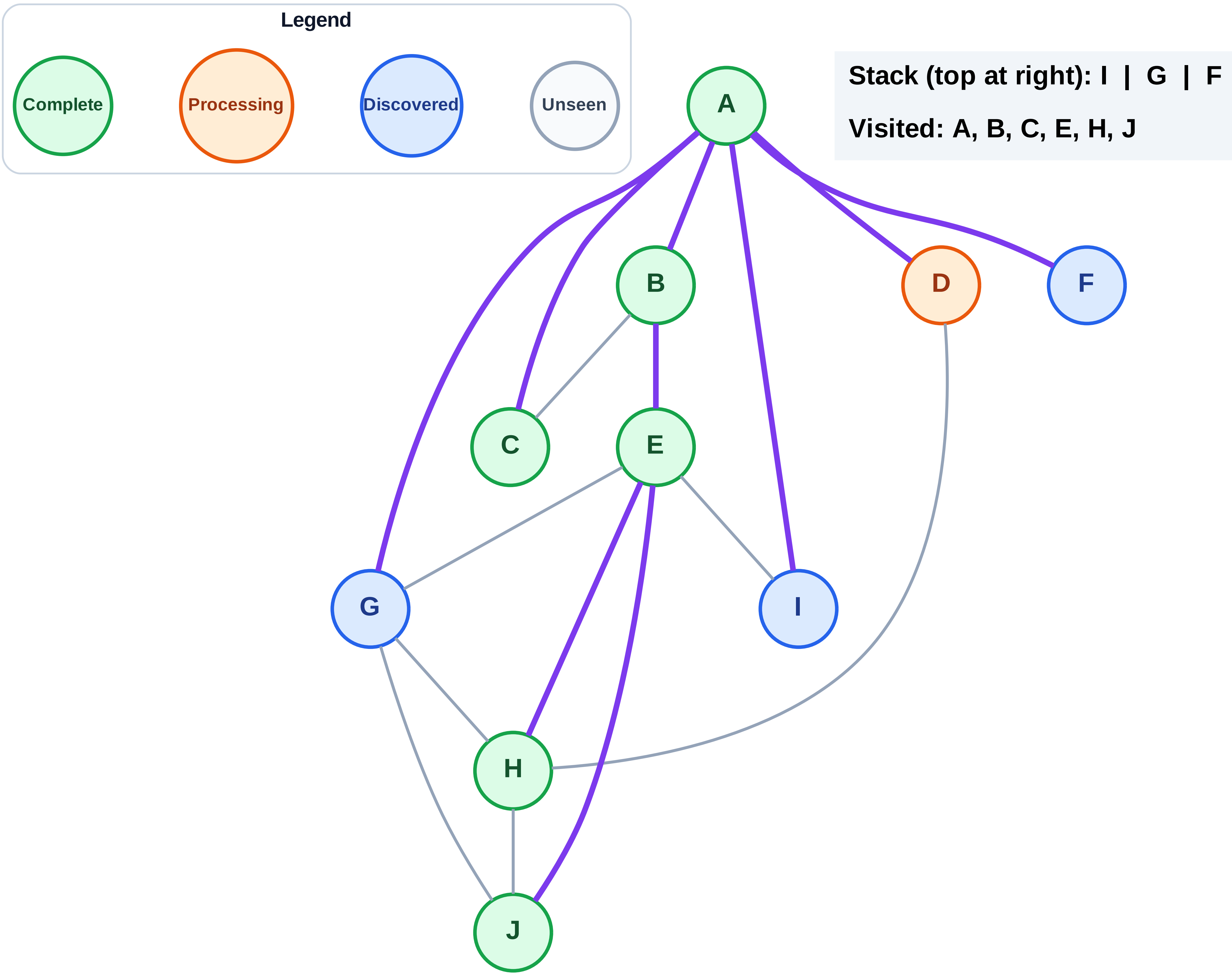
DFS Traversal

Step 13: No new neighbors from C



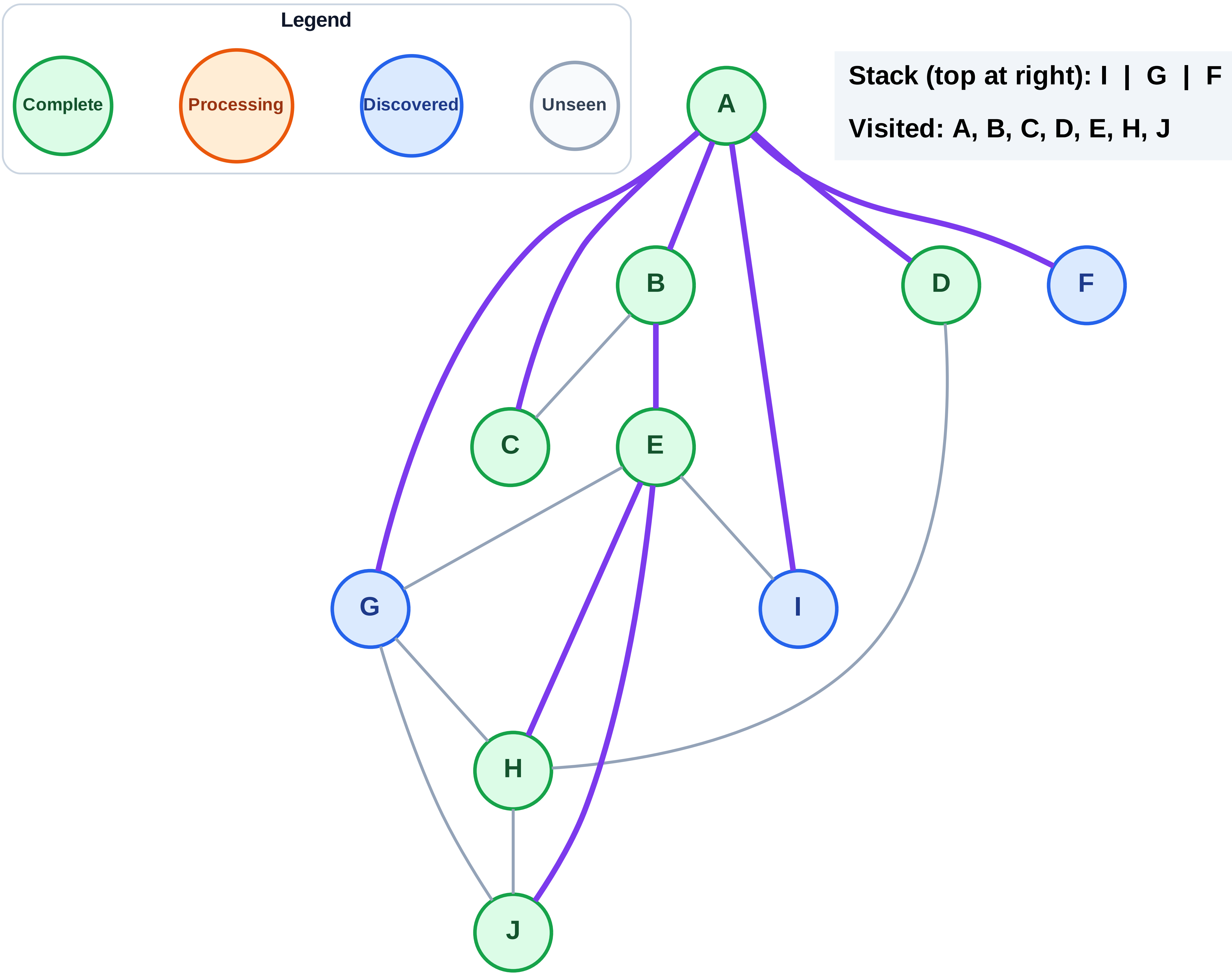
DFS Traversal

Step 14: Process node D



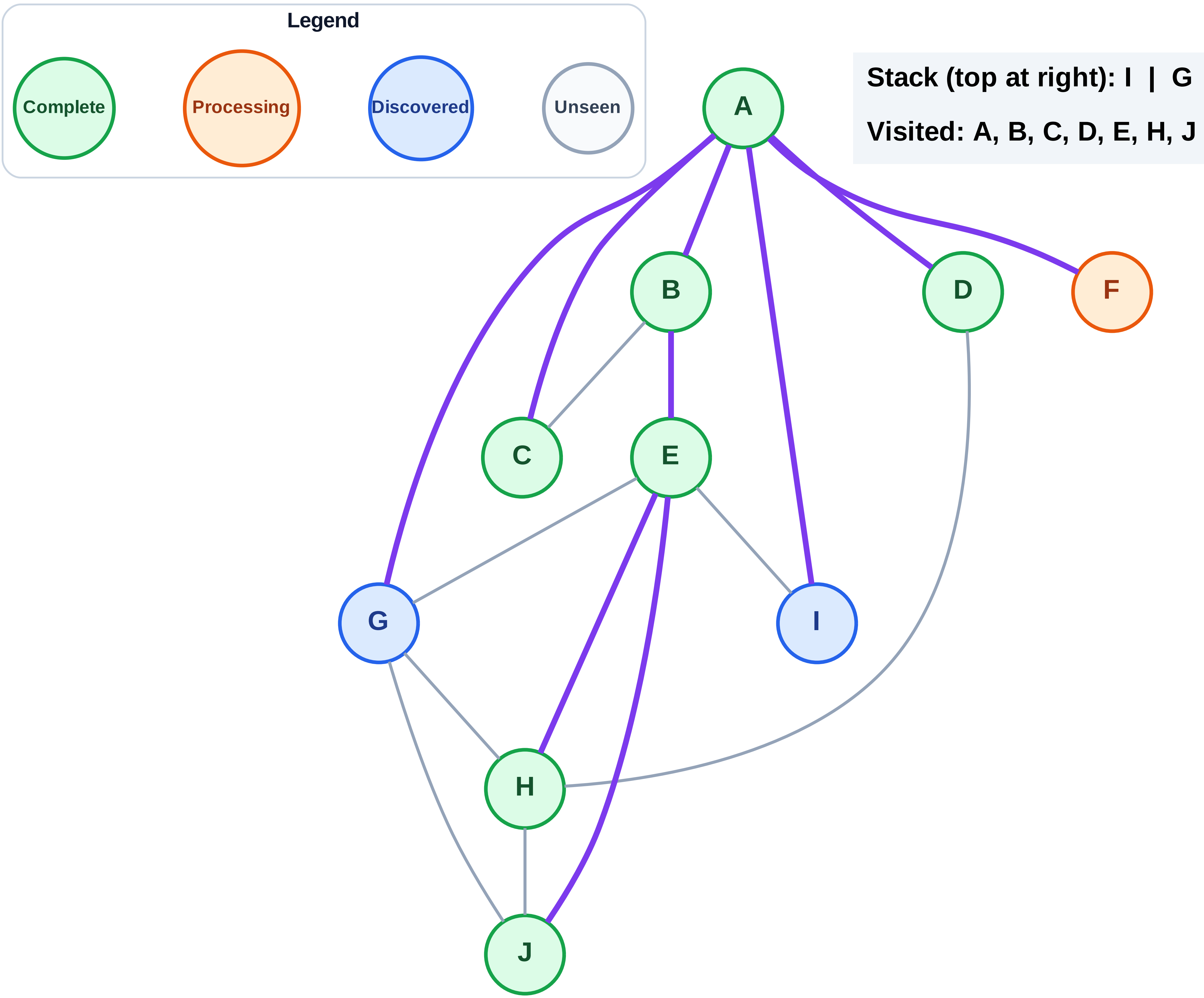
DFS Traversal

Step 15: No new neighbors from D



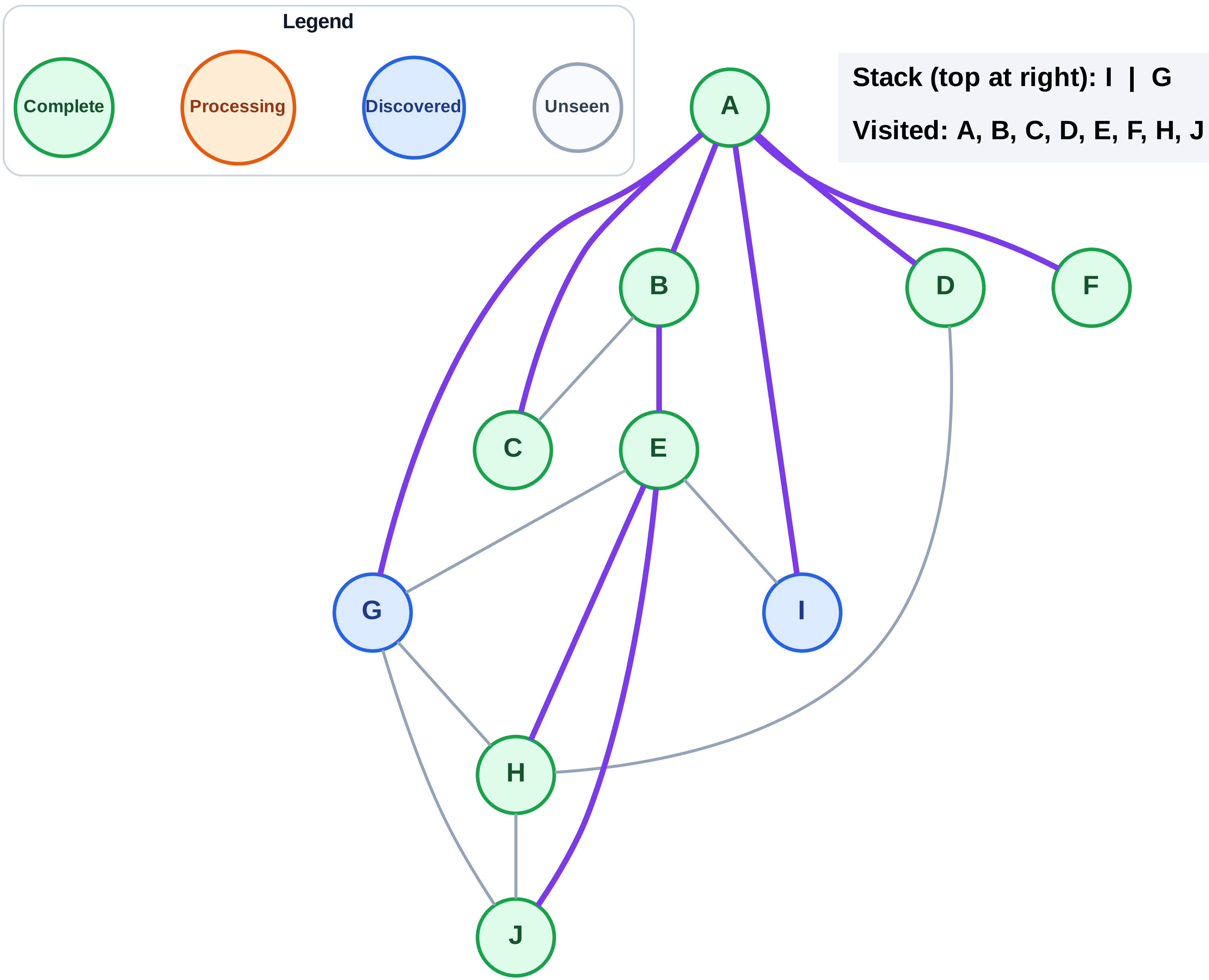
DFS Traversal

Step 16: Process node F



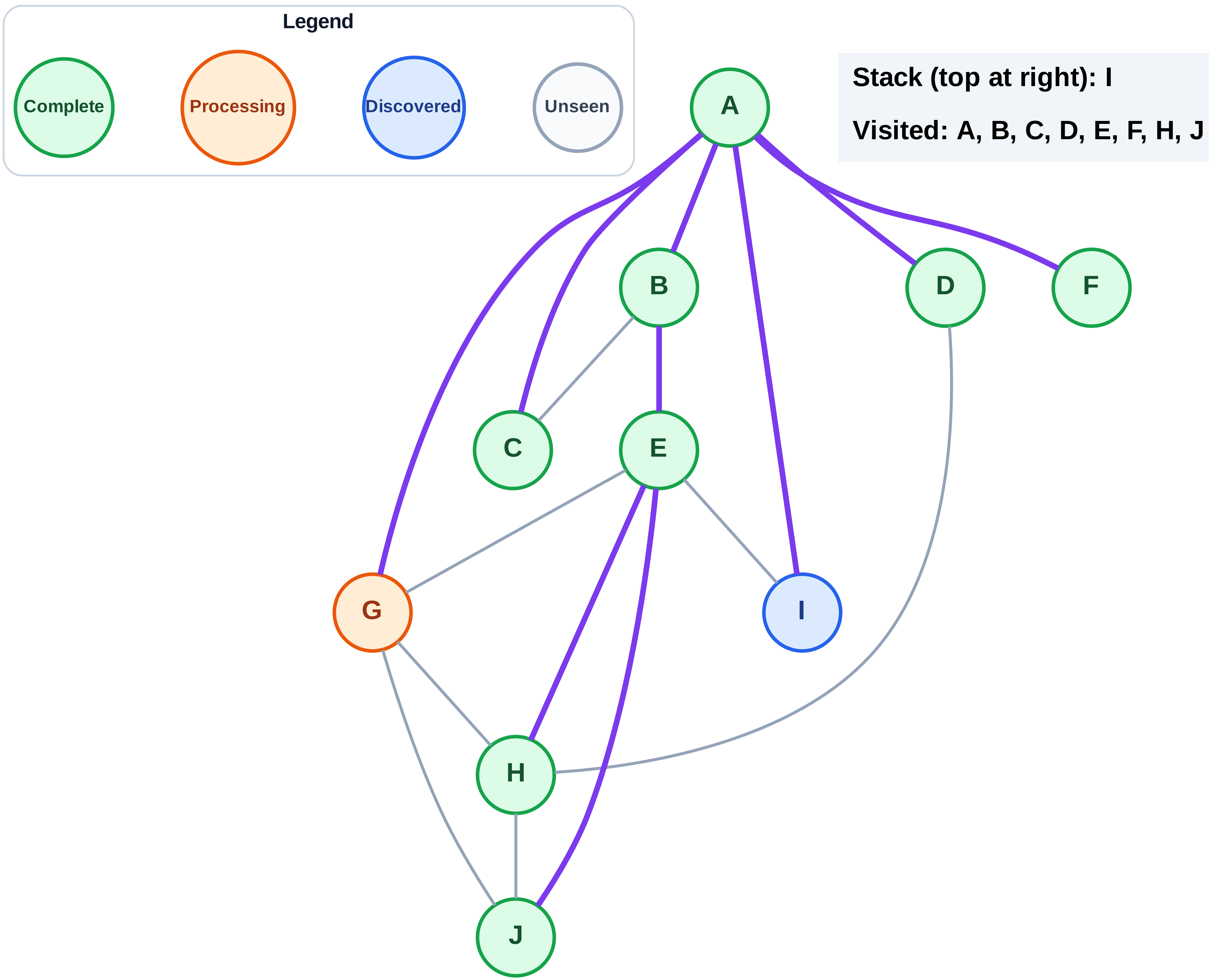
DFS Traversal

Step 17: No new neighbors from F



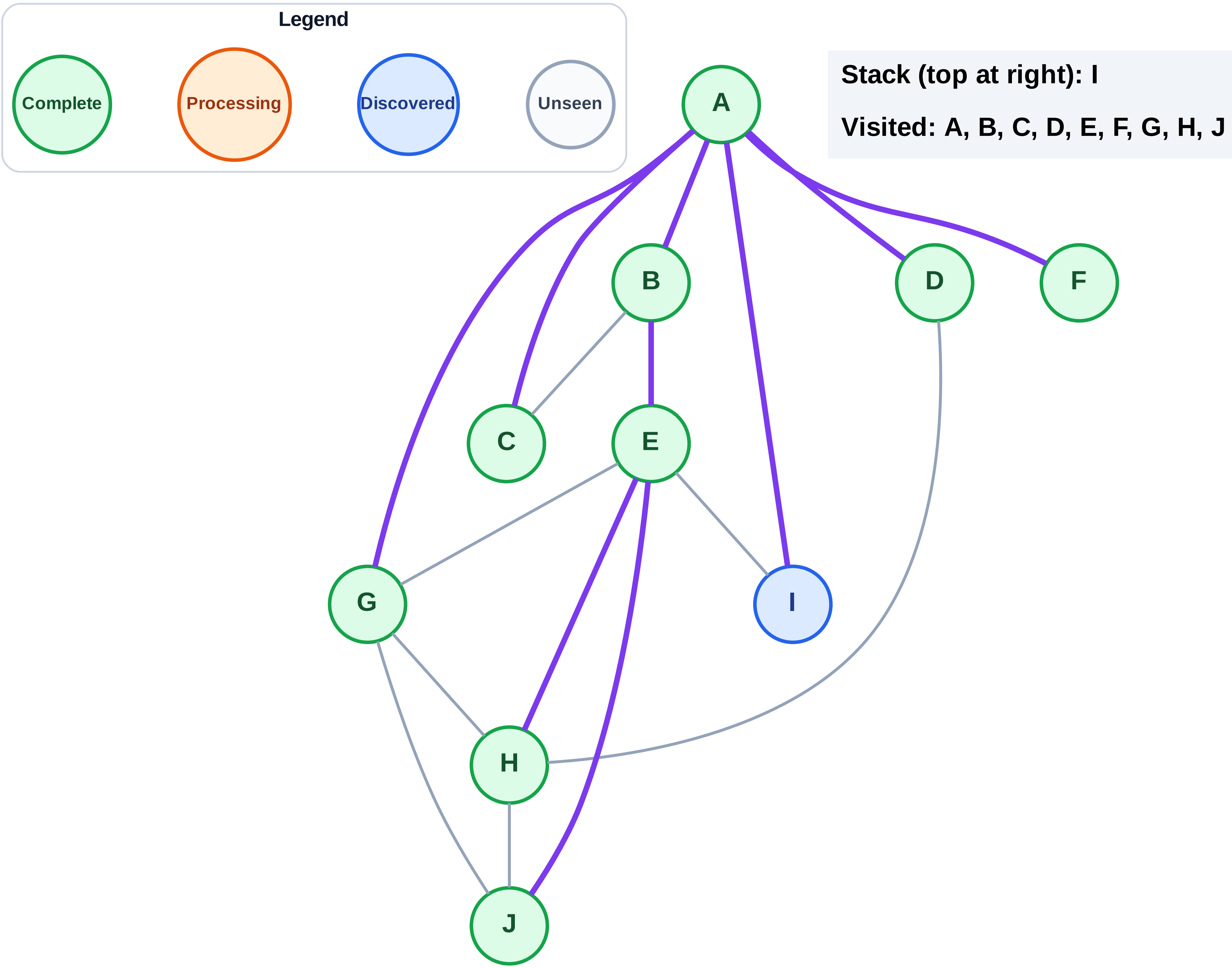
DFS Traversal

Step 18: Process node G



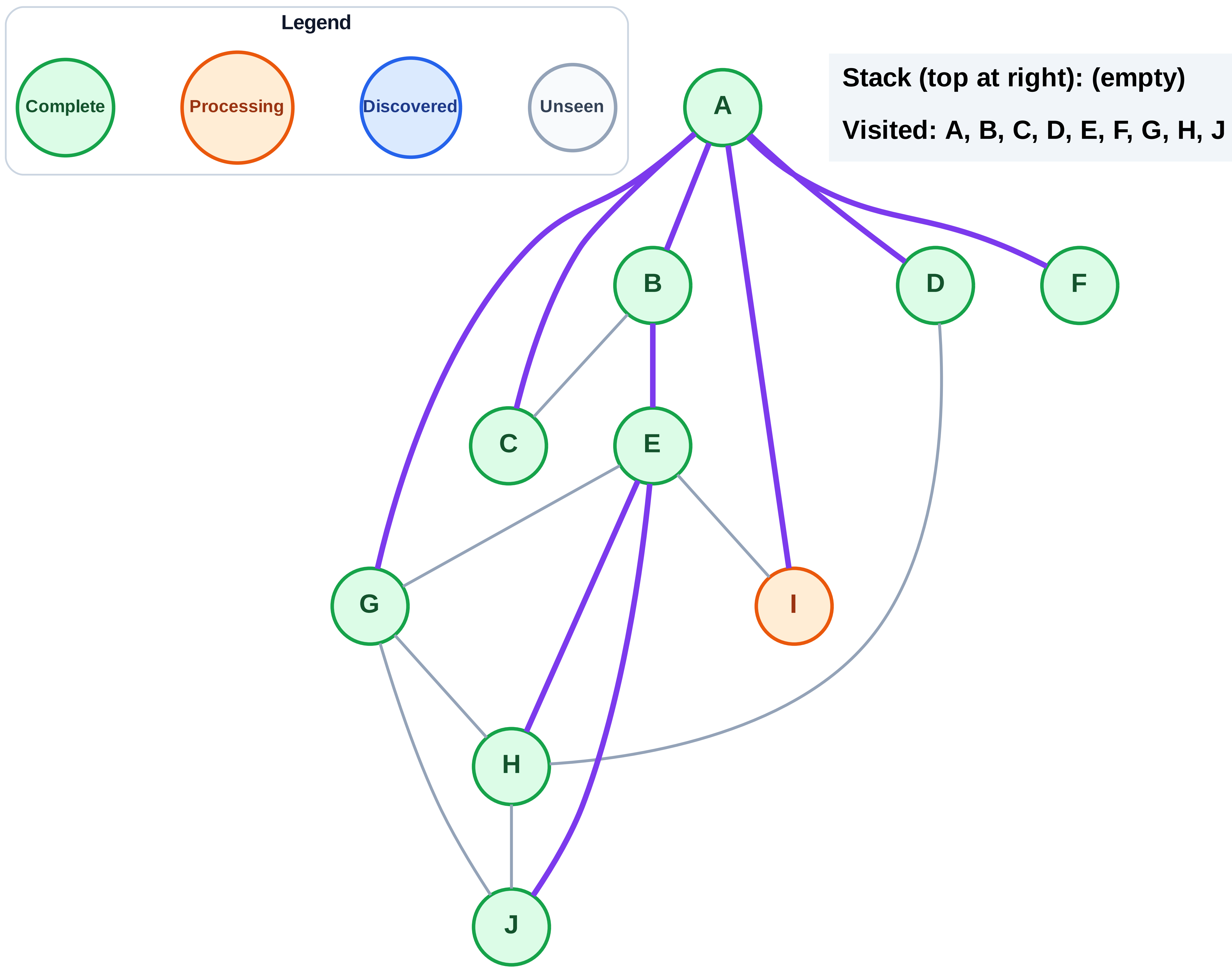
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node I



DFS Traversal

Step 21: No new neighbors from I